

Adiabatic Error Cancellation in Berry Phase Estimation

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In this work, we show that Berry phase estimation admits a natural and universal adiabatic error-cancellation mechanism, making it a promising candidate for practical quantum computing before full fault tolerance. Combining finite-runtime evolutions under $\pm H$ along the loop cancels the leading $O(T^{-1})$ phase error exactly, and Richardson extrapolation further reduces the residual error to an oscillatory term with endpoint-controlled coefficient $O(\|\dot{H}(0)\|^2 \Delta(0)^{-4} T^{-2})$. Beyond this deterministic cancellation, we establish that, for suitable smooth runtime distributions, runtime randomization suppresses the remaining oscillatory contribution to $O(T^{-M})$ for any fixed M , leading to a randomized Hadamard-test algorithm for Berry phase estimation over the full range $[0, 2\pi)$ with improved runtime scaling under standard sample complexity.

I. INTRODUCTION

Quantum computers can offer exponential speedups for certain computational tasks, with quantum simulation of many-body systems being one of their most compelling applications [2–4]. Recent advances in hardware have brought the field closer to early fault-tolerant quantum computation. In the NISQ (noisy intermediate-scale quantum) regime, devices operate without error correction. In the early fault-tolerant regime, limited error correction is available, but resources remain constrained. In both settings, identifying physically meaningful problems that are intrinsically robust to noise arising from quantum evolution is of central importance for achieving practical quantum advantage before full fault tolerance [5, 6].

Berry phase estimation—determining the geometric phase acquired by a quantum state transported adiabatically around a loop in parameter space [7, 8]—could be a promising problem in this context. Our motivation stems from the observation that the Berry phase, as a manifestation of quantum geometry, depends only on the path traced in parameter space, not on the speed of traversal or the energy spectrum. This suggests that Berry phase estimation may be inherently more robust to certain classes of errors than other quantum computational tasks such as energy estimation.

The Berry phase is one of the most fundamental concepts in modern quantum physics. Since its discovery, it has profoundly reshaped our understanding of quantum matter: it provides the theoretical foundation for the classification of topological phases of matter [9, 10], explains the modern theory of electric polarization in crystalline solids [11–13], underlies the anomalous Hall effect [14, 15], and captures the quantum geometry of Bloch states that governs superfluid weight and optical responses in flat-band systems [16, 17]. Its integration over closed surfaces yields

topological invariants such as the Chern number [18], while its non-abelian generalization—the Wilczek–Zee holonomy [19]—extends these ideas to degenerate subspaces and forms the basis of holonomic quantum computation [20, 21].

Estimating the Berry phase is therefore a problem of broad physical importance. Sophisticated classical methods exist [13, 22–24], but they become intractable for strongly correlated many-body systems in general. Indeed, Berry phase estimation has recently been shown to be BQP-complete with a known guiding state [1], establishing that quantum computers offer a provable advantage for this problem. Several proposals for estimating Berry phases on quantum computers have been put forward [1, 25–28], typically combining adiabatic simulation with phase estimation or interferometric techniques. However, a systematic analysis of the adiabatic phase error and its scaling in terms of the Hamiltonian parameters has been lacking, and no intrinsic robustness mechanism has been theoretically identified.

Since Berry phase estimation on a quantum computer requires simulating the adiabatic evolution of the ground state around a closed loop, the adiabatic error from finite-speed traversal is the most fundamental source of systematic error. The analysis of adiabatic error has a long history. At the foundational level, the adiabatic theorem was established by Born and Fock and reformulated in a modern projection-theoretic language by Kato [29, 30]. Berry developed an adiabatic-iteration scheme yielding successive corrections to the phase [31], while Aharonov and Anandan introduced the nonadiabatic geometric phase [32]. In parallel, rigorous adiabatic theory developed along a different direction, emphasizing spectral transport and leakage estimates; important milestones include the work of Avron, Seiler, and Yaffe [33], the exponential-error analysis of Nenciu [34], and the gap-explicit bounds of Jansen, Ruskai, and Seiler [35]. Rigolin, Ortiz, and Ponce developed a systematic adiabatic perturbation theory (APT) in powers of the inverse runtime [36], providing the algebraic framework for computing finite-speed corrections to adiabatic evolution.

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Method	Leading Phase Error	Runtime Scaling	Reference
Runtime scaling method [1]	$O(T^{-1})$, non-osc.	$O\left(\frac{\dot{H}_{\max}^2}{\Delta_{\min}^3 \varepsilon_B}\right)$	Lemma 1
+ Forward–reverse	$O(T^{-2})$, osc. + non-osc.	Appendix C	Theorem 1
+ Richardson extrapolation	$O(T^{-2})$, osc.	$O\left(\frac{\ \dot{H}(0)\ }{\Delta(0)^2 \varepsilon_B^{1/2}}\right)$	Theorem 2
+ Runtime randomization	$O(T^{-(M+2)})$, bias	$O\left(\frac{\ \dot{H}(0)\ ^{2/(M+2)}}{\Delta(0)^{4/(M+2)} \Delta_{\min}^{M/(M+2)} \varepsilon_B^{1/(M+2)}}\right)$	Theorem 3

TABLE I. Comparison of Berry phase estimation algorithms. For the runtime-randomized method, we consider the Hadamard-test implementation in Section VIA with sample complexity $O(\varepsilon_B^{-2})$; for the others, we consider the QPE-based protocol in Appendix E and ignore the QPE error, reporting only the Hamiltonian-simulation cost. For simplicity, we also assume $\dot{H}(0) = \dot{H}(1)$. All methods require the runtime-scaling technique of Ref. [1] for branch resolution to achieve estimation over the full range $[0, 2\pi)$; its cost is independent of ε_B and is subdominant for small ε_B , so we omit it from the comparison. The first row corresponds to the single-evolution phase error estimate (Lemma 1), whose leading non-oscillatory contribution is of order T^{-1} . The forward–reverse protocol cancels the leading $O(T^{-1})$ term and yields residual non-oscillatory and oscillatory contributions of order T^{-2} (Theorem 1). Richardson extrapolation removes the non-oscillatory part of this $O(T^{-2})$ residual, leaving only an oscillatory $O(T^{-2})$ term (Theorem 2). For suitable smooth runtime distributions and with recursive extrapolation, runtime randomization suppresses the remaining oscillatory contribution to arbitrarily high order in $1/T$; for any fixed positive integer M , the leading deterministic bias can be reduced to $O(T^{-(M+2)})$ (Theorem 3).

In this work, we derive an explicit large- T expansion of the adiabatic phase error in powers of $1/T$ using adiabatic perturbation theory [36] within the wave-operator framework of Ref. [35]. This reveals a natural and universal cancellation mechanism: combining the forward adiabatic evolution under H with the reverse evolution under $-H$ along the same loop cancels the leading $O(T^{-1})$ phase error exactly, improving the systematic error to $O(T^{-2})$. We further show that Richardson extrapolation removes the non-oscillatory part of the residual $O(T^{-2})$ term, yielding an endpoint-controlled error bound of order $O(\|\dot{H}(0)\|^2/(\Delta(0)^4 T^2))$. When incorporated into a quantum phase estimation algorithm, this cancellation improves the total cost from $O(\varepsilon_B^{-2})$ to $O(\varepsilon_B^{-3/2})$, where ε_B is the target accuracy. The mod- π ambiguity inherent to the forward–reverse protocol is resolved by a branch-resolution step based on the runtime-scaling technique of Ref. [1], whose cost is independent of ε_B , thereby enabling estimation over the full phase range $[0, 2\pi)$. Beyond this deterministic improvement, we show that uniform runtime randomization further suppresses the remaining oscillatory contribution. In a Hadamard-test implementation, averaging over uniformly randomized runtimes suppresses the oscillatory contribution by one additional factor of $1/T$ in expectation, so that the leading deterministic bias becomes $O(T^{-3})$. This leads to a randomized algorithm for Berry phase estimation over the full phase range $[0, 2\pi)$, with improved runtime scaling under standard sample complexity. Furthermore, by choosing runtime distributions with sufficiently rapidly decaying characteristic function, the oscillatory bias can be suppressed to $O(T^{-M})$ for any fixed M . Taken together, our results provide rigorous evidence that Berry

phase estimation is intrinsically more robust to finite-speed adiabatic error than energy estimation, and they point to Berry-phase estimation as a promising setting in which practical quantum advantage may emerge before fully fault-tolerant quantum computation becomes available.

The paper is organized as follows. In Section II, we introduce the wave-operator formalism, define the adiabatic phase error, and derive the large- T expansions of the phase and leakage errors in Lemma 1 and Proposition 1. In Section III, we show that combining the evolutions under H and $-H$ cancels the leading $O(T^{-1})$ phase error (Theorem 1), and in Section IIIB we analyze Richardson extrapolation, which yields the improved second-order bound (Theorem 2). In Section IV, we show that runtime randomization, combined with recursive extrapolation, further suppresses the remaining oscillatory contribution and prove the resulting error bound for the randomized estimator (Theorem 3). In Section VI, we present the Berry phase estimation algorithm, establish its correctness and complexity, and compare it with prior work. In particular, Appendix E presents the QPE-based protocol, while Section VIA presents the Hadamard-test version and shows that runtime randomization can be incorporated naturally in that setting, yielding improved runtime scaling under standard sample complexity. In Section VII, we discuss extensions to Chern numbers and non-Abelian holonomies, the relation to energy estimation, and possible routes toward practical quantum advantage. The appendices review adiabatic perturbation theory and provide detailed derivations of the phase-error expansion and the main results. Table I provides a roadmap of the main complexity improvements and their

corresponding algorithmic regimes.

II. ADIABATIC ERROR ON PHASE

A. Adiabatic Evolution Along the Loop

Let $H(s)$ be a smooth family of Hamiltonians for $s \in [0, 1]$, and assume that $H(0) = H(1)$. For each $s \in [0, 1]$, let $E_n(s)$ and $|n(s)\rangle$ denote the n -th instantaneous eigenvalue and corresponding eigenstate of $H(s)$, with the ground state denoted by $|\psi(s)\rangle = |0(s)\rangle$. Assume moreover that $|\psi(s)\rangle$ is non-degenerate for every $s \in [0, 1]$. The minimum spectral gap along the loop is then

$$\Delta_{\min} := \min_{s \in [0, 1]} (E_1(s) - E_0(s)).$$

For a runtime T , the Schrödinger equation

$$i \frac{d}{dt} |\Psi(t)\rangle = H\left(\frac{t}{T}\right) |\Psi(t)\rangle, \quad t \in [0, T],$$

in the rescaled variable $s := t/T \in [0, 1]$ becomes

$$i \frac{d}{ds} |\Psi(s)\rangle = TH(s) |\Psi(s)\rangle.$$

The time-evolution operator up to rescaled time s is

$$U_T(s) = \mathcal{T} \exp\left(-iT \int_0^s H(\sigma) d\sigma\right),$$

where $\mathcal{T}$ denotes time ordering, so that

$$|\Psi(s)\rangle = U_T(s) |\Psi(0)\rangle.$$

We define the instantaneous dynamical phase by

$$\theta_D(s) := T \int_0^s E_0(\sigma) d\sigma$$

and the instantaneous Berry phase by

$$\theta_B(s) := \int_0^s i \langle \psi(\sigma) | \dot{\psi}(\sigma) \rangle d\sigma.$$

At the end of the loop, $\theta_D := \theta_D(1)$ and $\theta_B := \theta_B(1)$ are the dynamical and Berry phases, respectively.

The adiabatic theorem states that if the initial state is the ground state $|\psi(0)\rangle$, then the evolved state $U_T(s) |\psi(0)\rangle$ remains close to the instantaneous ground state $|\psi(s)\rangle$ for all $s \in [0, 1]$, and in particular acquires the dynamical and Berry phases up to an error of order $O(T^{-1})$:

$$U_T(s) |\psi(0)\rangle = e^{-i\theta_D(s)} e^{i\theta_B(s)} |\psi(s)\rangle + O(T^{-1}).$$

In particular, at the end of the loop,

$$U_T(1) |\psi(0)\rangle = e^{-i\theta_D} e^{i\theta_B} |\psi(0)\rangle + O(T^{-1}).$$

At finite T we quantify the deviation from this ideal transport by the scalar adiabatic error amplitude

$$z(s) := e^{i\theta_D(s)} e^{-i\theta_B(s)} \langle \psi(s) | U_T(s) | \psi(0) \rangle,$$

the overlap of the true evolution with the instantaneous ground state, with the exact dynamical and Berry phases removed. Its modulus quantifies the overlap with ideal adiabatic transport, while its argument measures the residual phase mismatch. Accordingly, we define the leakage probability and phase error by

$$p_{\text{leak}}(s) := 1 - |z(s)|^2, \quad \varphi(s) := \arg z(s), \quad (1)$$

where $p_{\text{leak}}(s)$ is the nonadiabatic transition probability out of the instantaneous ground state. Here we assume that $z(s) \neq 0$ for all $s \in [0, 1]$, so that $\varphi(s)$ is well defined.

We write $z(s) = 1 + \delta(s)$ with $\delta(s) := z(s) - 1$. By the adiabatic theorem the true evolution converges to ideal transport as $T \rightarrow \infty$, so $\delta(s) = O(T^{-1})$ and $|\delta(s)| < 1$ for sufficiently large T . The phase error is then the imaginary part of the principal logarithm of $z(s) = 1 + \delta(s)$, which admits the convergent expansion

$$\begin{aligned} \varphi(s) = \arg z(s) &= \Im \log(1 + \delta(s)) \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \Im(\delta(s)^k). \end{aligned} \quad (2)$$

Since $\delta(s) = O(T^{-1})$, the k -th term is $O(T^{-k})$, so to leading order the phase error is determined by the imaginary part of the first-order correction $\delta(s)$. The following theorem gives the resulting asymptotic formula.

B. Phase error expansion

We show the phase error expansion for a single adiabatic evolution, making the order-by-order structure of the error explicit. The proof, based on the adiabatic perturbation theory reviewed in Appendix A, is given in Appendix B. For notational simplicity, in addition to assuming that the ground state $|\psi(s)\rangle$ is non-degenerate for every $s \in [0, 1]$, we further assume throughout the remainder of this paper that all excited-state eigenvalues $E_n(s)$ with $n \neq 0$ are also non-degenerate, so that the entire instantaneous spectrum is fully non-degenerate for all $s \in [0, 1]$. We note, however, that since the initial state in our setting is the non-degenerate ground state, the underlying argument is expected to extend to the case where only the ground state remains separated from the excited manifold by a nonzero gap, even if degeneracies or level crossings occur among the excited states, as already suggested in the original APT work [36]. Since incorporating that level of generality would substantially complicate the notation without affecting the main ideas, we restrict ourselves here to the fully non-degenerate case.

Throughout the remainder of this paper, we denote the operator norm of the Hamiltonian by $H_{\max} :=$

$\max_s \|H(s)\|$, and the maximum norms of its first and second derivatives by

$$\dot{H}_{\max} := \max_s \|\dot{H}(s)\|, \quad \ddot{H}_{\max} := \max_s \|\ddot{H}(s)\|.$$

We frequently use the notation and standard identity obtained by differentiating the instantaneous eigenvalue equation,

$$M_n(s) := M_{n0}(s) = \frac{\langle n(s) | \dot{H}(s) | \psi(s) \rangle}{\Delta_n(s)}, \quad n \neq 0,$$

where $M_{nm}(s) := \langle n(s) | \dot{m}(s) \rangle$ is the adiabatic coupling matrix element and $\Delta_n(s) := \Delta_{n0}(s) := E_n(s) - E_0(s)$ is the spectral gap to the n -th level. This gives the uniform bound $|M_n(s)| \leq \dot{H}_{\max}/\Delta_{\min}$. We also denote the Berry phase of the n -th eigenstate by $\theta_B^{(n)}(s) := \int_0^s i \langle n(\sigma) | \dot{n}(\sigma) \rangle d\sigma$, and the relative Berry phase between the n -th level and the ground state by $\beta_n(s) := \theta_B^{(n)}(s) - \theta_B(s)$.

Lemma 1 (Phase error expansion). *Under the setting of Section II, for any fixed integer $K \geq 2$, the phase error admits the asymptotic expansion*

$$\varphi = \frac{\varphi_1}{T} + \sum_{k=2}^K \frac{\varphi_k + \varphi_k^{(T)}}{T^k} + O(T^{-K-1}). \quad (3)$$

Here φ_k are T -independent non-oscillatory loop contributions. For each $k \geq 2$, the oscillatory boundary coefficient $\varphi_k^{(T)}$ is a finite linear combination of sine and cosine modes with nonzero frequencies:

$$\varphi_k^{(T)} = \sum_{\nu \in I_k} [a_{k,\nu} \cos(\Omega_{k,\nu} T) + b_{k,\nu} \sin(\Omega_{k,\nu} T)]. \quad (4)$$

Here I_k is finite, $a_{k,\nu}$ and $b_{k,\nu}$ are independent of T , and each $\Omega_{k,\nu}$ is a positive sum of integrated gaps $\omega_{n0}(1)$. Hence $\Omega_{k,\nu} \geq \Delta_{\min} > 0$, so no zero-frequency component appears in the oscillatory sector. Here and below, all quantities depend on s unless evaluated at a specific endpoint. The leading coefficients are

$$\begin{aligned} \varphi_1 &= \int_0^1 \sum_{n \neq 0} \frac{|M_n|^2}{\Delta_n} ds, \\ \varphi_2 &= \sum_{n \neq 0} \int_0^1 \frac{(\dot{\beta}_n - \dot{\phi}_n) |M_n|^2}{\Delta_n^2} ds - i \int_0^1 \mathcal{A}^{(2)} ds, \\ \varphi_2^{(T)} &= - \sum_{n \neq 0} \frac{|M_n(0)| |M_n(1)|}{\Delta_n(0)^2} \\ &\quad \times \sin(\beta_n(1) + \phi_n(0) - \phi_n(1) - \omega_n T), \end{aligned}$$

where

$$\phi_n(s) := \arg M_n(s), \quad \omega_n := \int_0^1 \Delta_n(\sigma) d\sigma,$$

and

$$\mathcal{A}^{(2)} := \sum_{\substack{n,k \neq 0 \\ n \neq k}} \frac{M_{0n} M_{nk} M_{k0}}{\Delta_n \Delta_k} \quad (5)$$

is purely imaginary, so that φ_2 is real. Moreover, the first-order coefficient and the leading oscillatory coefficient obey

$$\varphi_1 \leq \frac{\dot{H}_{\max}^2}{\Delta_{\min}^3}, \quad |\varphi_2^{(T)}| \leq \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4},$$

where $\Delta(0) := \Delta_1(0) = E_1(0) - E_0(0)$.

A bound on φ_2 is given in Appendix C; the bounds on φ_1 and $\varphi_2^{(T)}$ stated in Lemma 1 follow from the completeness of the eigenbasis. If in addition $\dot{H}(0) = \dot{H}(1)$, then the closed-loop condition $H(1) = H(0)$ gives $M_n(0) = M_n(1)$ and $\phi_n(0) = \phi_n(1)$, and $\varphi_2^{(T)}$ simplifies to

$$\varphi_2^{(T)} = - \sum_{n \neq 0} \frac{|M_n(0)|^2}{\Delta_n(0)^2} \sin(\beta_n(1) - \omega_n T) \leq \frac{\|\dot{H}(0)\|^2}{\Delta(0)^4}.$$

In contrast to the $\Delta_{\min}^{-3}$ scaling of φ_1 , the bound on $\varphi_2^{(T)}$ is endpoint-controlled.

The three contributions to the phase error have distinct origins. The first-order coefficient φ_1 is a T -independent, gauge-invariant quantity that measures the cumulative coupling between the ground state and the excited states along the loop, weighted by the inverse spectral gaps. The non-oscillatory second-order coefficient φ_2 consists of two parts: an integral involving the mismatch between the relative Berry-phase rate $\dot{\beta}_n$ and the coupling-phase rate $\dot{\phi}_n$, and the integral of $\mathcal{A}^{(2)}$, which captures virtual excitation processes $|0\rangle \rightarrow |n\rangle \rightarrow |k\rangle \rightarrow |0\rangle$ weighted by the spectral gaps. Both φ_1 and φ_2 are T -independent, gauge-invariant coefficients determined by the loop. By contrast, the oscillatory term $\varphi_2^{(T)}$ depends explicitly on T through $\sin(\dots + \omega_n T)$ and arises from boundary contributions at $s = 0$ and $s = 1$ that do not cancel under the loop condition.

C. Leakage error expansion

In addition to the phase error within the adiabatically transported ground-state sector, it is also important to quantify the probability of nonadiabatic transitions out of that sector. For this purpose, we focus on the leakage probability $p_{\text{leak}}(s)$ defined in Eq. (1). Thus the leakage probability measures the loss of weight from the transported ground-state sector, whereas the phase error $\varphi(s) = \arg z(s)$ measures the argument of the same survival amplitude.

Unlike the phase error, the leading-order leakage probability is obtained directly from the first-order excited-state amplitudes in adiabatic perturbation theory. The following proposition follows from a direct computation of the first-order correction derived in Appendix A.

Proposition 1 (Leakage error expansion). *Under the setting of Section II, the leakage probability satisfies*

$$p_{\text{leak}}(1) = \frac{1}{T^2} \sum_{n \neq 0} \left| \frac{M_n(1)}{\Delta_n(1)} - e^{-iT\omega_n} e^{i\beta_n(1)} \frac{M_n(0)}{\Delta_n(0)} \right|^2 + O(T^{-3}). \quad (6)$$

In particular,

$$p_{\text{leak}}(1) \leq \frac{C_{\text{leak}}}{T^2} + O(T^{-3}),$$

$$C_{\text{leak}} = O\left(\frac{\|\dot{H}(0)\|^2 + \|\dot{H}(1)\|^2}{\Delta(0)^4}\right).$$

where $\Delta(0)$ denotes the minimum energy gap at $s = 0$, i.e., $\Delta(0) := \Delta_1(0) = E_1(0) - E_0(0)$.

If, in addition, $\dot{H}(0) = \dot{H}(1)$, then $M_{n0}(1) = M_{n0}(0)$, and Eq. (6) reduces to

$$p_{\text{leak}}(1) = \frac{4}{T^2} \sum_{n \neq 0} \frac{|M_n(0)|^2}{\Delta_n(0)^2} \sin^2\left(\frac{\omega_n T - \beta_n(1)}{2}\right) + O(T^{-3}).$$

Thus, we obtain

$$p_{\text{leak}}(1) \leq \frac{C}{T^2} + O(T^{-3}), \quad C = O\left(\frac{\|\dot{H}(0)\|^2}{\Delta(0)^4}\right). \quad (7)$$

Compared with the phase error expansion, the leakage formula is much simpler: it is determined solely by the squared modulus of the first-order excited-state amplitudes, and therefore starts at order T^{-2} . By contrast, the phase error of a single adiabatic evolution is generically $O(T^{-1})$. This distinction becomes important below: after the forward–reverse cancellation suppresses the $O(T^{-1})$ phase bias, the residual phase error and the leakage probability are both $O(T^{-2})$, so their relative importance is set by their coefficients rather than by the runtime scaling alone.

III. ADIABATIC ERROR CANCELLATION

A. First-order Cancellation

To isolate the Berry phase while canceling the dynamical phase, we introduce the reverse evolution generated by $-H(s)$:

$$\hat{U}_T(s) := \mathcal{T} \exp\left(+iT \int_0^s H(\sigma) d\sigma\right). \quad (8)$$

Since $-H(s)$ has eigenvalues $-E_n(s)$ with the same eigenstates $|n(s)\rangle$, the same instantaneous eigenstate $|\psi(s)\rangle$ is followed adiabatically. Hence the ideal reverse

phase is obtained by changing the sign of the dynamical phase while leaving the Berry phase unchanged:

$$\hat{U}_T(s)|\psi(0)\rangle = e^{+i\theta_D(s)} e^{i\theta_B(s)} |\psi(s)\rangle + O(T^{-1}).$$

We therefore define the reverse adiabatic error amplitude by

$$\hat{\varphi}(s) := \arg\left(e^{-i\theta_D(s)} e^{-i\theta_B(s)} \langle\psi(s)|\hat{U}_T(s)|\psi(0)\rangle\right). \quad (9)$$

Since $-H(s)$ satisfies the same loop conditions as $H(s)$, Lemma 1 applies with the substitution $\Delta_n \rightarrow -\Delta_n$ (while M_n , β_n , ϕ_n , and $\mathcal{A}^{(2)}$ remain unchanged), giving

$$\hat{\varphi} = -\frac{\varphi_1}{T} + \frac{\varphi_2}{T^2} + \frac{\varphi_2^{(-T)}}{T^2} + O(T^{-3}), \quad (10)$$

where the reverse evolution is formally obtained by the substitution $T \rightarrow -T$. Accordingly, the $O(T^{-1})$ term changes sign, the non-oscillatory $O(T^{-2})$ term remains unchanged, and the oscillatory contribution $\varphi_2^{(T)}$ is replaced by $\varphi_2^{(-T)}$.

When the eigenphases of $U_T(1)$ and $\hat{U}_T(1)$ are obtained independently, the forward eigenphase is $-\theta_D + \theta_B + \varphi$ and the reverse eigenphase is $+\theta_D + \theta_B + \hat{\varphi}$. Averaging the two, the dynamical phases $\pm\theta_D$ cancel exactly, yielding the estimated Berry phase

$$\tilde{\theta}_B(T) := \theta_B + \frac{\varphi + \hat{\varphi}}{2}. \quad (11)$$

Note that $\tilde{\theta}_B(T)$ is determined only modulo π , since the sum of the two eigenphases yields $2\theta_B + \varphi + \hat{\varphi}$ and dividing by two introduces a π -ambiguity. This restriction is resolved by a branch-resolution step in Section VI.

The following theorem shows that the leading-order term cancels, leaving an $\tilde{\theta}_B(T) - \theta_B = O(T^{-2})$ error, which improves on the $O(T^{-1})$ error of a single evolution. The proof is given in Appendix B.

Theorem 1 (Odd order adiabatic error cancellation). *Under the setting of Section II, the forward–reverse symmetrized estimator cancels the non-oscillatory terms*

$$\frac{\Phi_1}{T}, \frac{\Phi_3}{T^3}, \frac{\Phi_5}{T^5}, \dots$$

exactly. More precisely, for any fixed integer $K \geq 2$, the estimated Berry phase satisfies

$$\tilde{\theta}_B(T) - \theta_B = \sum_{k=1}^{\lfloor K/2 \rfloor} \frac{\Phi_{2k}}{T^{2k}} + \sum_{k=2}^K \frac{\Phi_k^{(T)}}{T^k} + O(T^{-K-1}), \quad (12)$$

Here $\Phi_{2k} = \varphi_{2k}$ are non-oscillatory coefficients, while $\Phi_k^{(T)} = (\varphi_k^{(T)} + \varphi_k^{(-T)})/2$ are oscillatory coefficients depending on T and

$$\Phi_2^{(T)} = -\sum_{n \neq 0} B_n \cos(\omega_n T), \quad (13)$$

where

$$B_n := \frac{|M_n(0)| |M_n(1)|}{\Delta_n(0)^2} \sin(\beta_n(1) + \phi_n(0) - \phi_n(1)). \quad (14)$$

If in addition $\dot{H}(0) = \dot{H}(1)$, then $M_n(0) = M_n(1)$ and $\phi_n(0) = \phi_n(1)$, and the oscillatory term simplifies to

$$\Phi_2^{(T)} = - \sum_{n \neq 0} \frac{|M_n(0)|^2}{\Delta_n(0)^2} \sin(\beta_n(1)) \cos(\omega_n T).$$

The remaining error consists of non-oscillatory even-order terms and oscillatory terms. In the next subsection, we apply Richardson extrapolation to remove the leading non-oscillatory $O(T^{-2})$ term.

B. Higher-order non-oscillatory cancellation

The residual error established in Theorem 1 has the following finite-order asymptotic form. For any fixed integer $K \geq 2$,

$$\tilde{\theta}_B(T) - \theta_B = \sum_{k=1}^{\lfloor K/2 \rfloor} \frac{\Phi_{2k}}{T^{2k}} + \sum_{k=2}^K \frac{\Phi_k^{(T)}}{T^k} + O(T^{-K-1}). \quad (15)$$

Here Φ_{2k} are T -independent non-oscillatory contributions, whereas $\Phi_k^{(T)} = O(1)$ are bounded rapidly oscillatory boundary terms. The leading non-oscillatory contribution is Φ_2/T^2 , and the leading oscillatory contribution is

$$\Phi_2^{(T)} = - \sum_{n \neq 0} B_n \cos(\omega_n T). \quad (16)$$

We now remove the non-oscillatory even-order terms by recursive Richardson extrapolation.

Let $m \geq 1$ be fixed. We define the m -fold Richardson extrapolant by

$$\tilde{\theta}_{B,R}^{(m)}(T) := \sum_{\ell=0}^m w_{m,\ell} \tilde{\theta}_B(\alpha^\ell T), \quad \alpha > 1. \quad (17)$$

The weights are chosen so that

$$\sum_{\ell=0}^m w_{m,\ell} = 1, \quad \sum_{\ell=0}^m w_{m,\ell} \alpha^{-2r\ell} = 0, \quad r = 1, \dots, m. \quad (18)$$

Equivalently, setting $x_\ell = \alpha^{-2\ell}$, the weights are

$$w_{m,\ell} = \prod_{q \in Q_{m,\ell}} \frac{-x_q}{x_\ell - x_q}, \quad Q_{m,\ell} := \{0, \dots, m\} \setminus \{\ell\}. \quad (19)$$

Substituting Eq. (15) into Eq. (17), we obtain

$$\tilde{\theta}_{B,R}^{(m)}(T) - \theta_B = \sum_{\substack{2r \leq K \\ r \geq m+1}} \frac{C_{r,m} \Phi_{2r}}{T^{2r}} + \sum_{k=2}^K \frac{\Phi_{k,m}^{(T)}}{T^k} + O(T^{-K-1}), \quad (20)$$

where

$$C_{r,m} := \sum_{\ell=0}^m w_{m,\ell} \alpha^{-2r\ell}, \quad (21)$$

and

$$\Phi_{k,m}^{(T)} := \sum_{\ell=0}^m w_{m,\ell} \alpha^{-k\ell} \Phi_k^{(\alpha^\ell T)}. \quad (22)$$

Thus the non-oscillatory terms

$$\frac{\Phi_2}{T^2}, \frac{\Phi_4}{T^4}, \dots, \frac{\Phi_{2m}}{T^{2m}}$$

cancel exactly. The oscillatory sector is not canceled by Richardson extrapolation; it is transformed into another bounded rapidly oscillatory function of T .

Applying the Richardson weights to the oscillatory modes gives

$$\Phi_{k,m}^{(T)} = \sum_{\nu \in I_k} \sum_{\ell=0}^m w_{m,\ell} \alpha^{-k\ell} S_{k,\nu}^{(\ell)}(T), \quad (23)$$

where

$$S_{k,\nu}^{(\ell)}(T) := a_{k,\nu} \cos(\alpha^\ell \Omega_{k,\nu} T) + b_{k,\nu} \sin(\alpha^\ell \Omega_{k,\nu} T). \quad (24)$$

In particular, the leading remaining oscillatory coefficient is

$$\Phi_{2,m}^{(T)} = - \sum_{n \neq 0} B_n \sum_{\ell=0}^m w_{m,\ell} \alpha^{-2\ell} \cos(\alpha^\ell \omega_n T). \quad (25)$$

Hence

$$\tilde{\theta}_{B,R}^{(m)}(T) - \theta_B = \frac{\Phi_{2,m}^{(T)}}{T^2} + O_{\text{osc}}(T^{-3}) + O_{\text{nonosc}}(T^{-2m-2}). \quad (26)$$

Here $O_{\text{osc}}(T^{-3})$ collects the higher-order oscillatory boundary terms, while $O_{\text{nonosc}}(T^{-2m-2})$ denotes the first possible remaining T -independent non-oscillatory contribution.

Theorem 2 (Higher-order non-oscillatory adiabatic error cancellation). *Under the setting of Section II, the m -fold Richardson extrapolant $\tilde{\theta}_{B,R}^{(m)}(T)$ defined in Eq. (17) cancels the non-oscillatory terms*

$$\frac{\Phi_2}{T^2}, \frac{\Phi_4}{T^4}, \dots, \frac{\Phi_{2m}}{T^{2m}}$$

exactly. More precisely, for any fixed integer $K \geq 2m+2$,

$$\tilde{\theta}_{B,R}^{(m)}(T) - \theta_B = \sum_{k=2}^K \frac{\Phi_{k,m}^{(T)}}{T^k} + \sum_{j=m+1}^{\lfloor K/2 \rfloor} \frac{C_{j,m} \Phi_{2j}}{T^{2j}} + O(T^{-K-1}). \quad (27)$$

Here $\Phi_{k,m}^{(T)}$ is defined in Eq. (23). The second sum is the remaining T -independent non-oscillatory sector, with $C_{j,m}$ defined in Eq. (21). In particular, the first possible non-oscillatory term is of order T^{-2m-2} .

For $m = 1$, Eq. (17) reduces to the usual second-order Richardson extrapolant

$$\tilde{\theta}_{B,R}^{(1)}(T) = \frac{\alpha^2 \tilde{\theta}_B(\alpha T) - \tilde{\theta}_B(T)}{\alpha^2 - 1}. \quad (28)$$

The leading oscillatory remainder becomes

$$\Phi_{2,1}^{(T)} = \frac{1}{\alpha^2 - 1} \sum_{n \neq 0} B_n [\cos(\omega_n T) - \cos(\alpha \omega_n T)]. \quad (29)$$

The size of the leading oscillatory coefficient is controlled by the same endpoint quantity as before:

$$\begin{aligned} |\Phi_{2,m}^{(T)}| &\leq L_m(\alpha) \sum_{n \neq 0} |B_n| \\ &\leq L_m(\alpha) \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4}. \end{aligned} \quad (30)$$

where

$$L_m(\alpha) := \sum_{\ell=0}^m |w_{m,\ell}| \alpha^{-2\ell}. \quad (31)$$

For fixed $\alpha > 1$, the amplification factor obeys

$$L_m(\alpha) \leq (m+1) \left(\frac{\alpha^2}{\alpha^2 - 1} \right)^m. \quad (32)$$

Thus higher-order Richardson extrapolation systematically removes the non-oscillatory even-order sector, but the deterministic prefactor grows with the extrapolation order. It is therefore natural to keep m fixed and small.

After m -fold Richardson extrapolation, the leading remaining term is a linear combination of oscillatory factors $\cos(\alpha^\ell \omega_n T)$. These terms are not suppressed by Richardson extrapolation alone. In the next section, we show that runtime randomization suppresses this oscillatory sector through the decay of the characteristic function of the runtime distribution. Thus Richardson removes the non-oscillatory sector, while runtime averaging suppresses the remaining oscillatory sector.

IV. ADIABATIC ERROR SUPPRESSION BY RUNTIME RANDOMIZATION

We now consider runtime randomization applied after Richardson extrapolation. To suppress the remaining oscillatory finite-time contributions, we randomize the base runtime according to a general distribution. Let $X_1, \dots, X_N$ be sampled independently from a probability distribution μ supported on $[1-\lambda, 1+\lambda]$, where $0 < \lambda < 1$, and define

$$T_j := T X_j.$$

For each realization T_j , we form the m -fold Richardson extrapolated estimator

$$\tilde{\theta}_{B,R}^{(m)}(T_j) := \sum_{\ell=0}^m w_{m,\ell} \tilde{\theta}_B(\alpha^\ell T_j).$$

Using the expansion of $\tilde{\theta}_B(T)$, the non-oscillatory terms

$$\frac{\Phi_2}{T_j^2}, \frac{\Phi_4}{T_j^4}, \dots, \frac{\Phi_{2m}}{T_j^{2m}}$$

cancel exactly, and the resulting single-event error becomes

$$\tilde{\theta}_{B,R}^{(m)}(T_j) - \theta_B = \sum_{k=2}^K \frac{\Phi_{k,m}^{(T_j)}}{T_j^k} + O_{\text{nonosc}}(T_j^{-2m-2}) + O(T_j^{-K-1}), \quad (33)$$

where $\Phi_{k,m}^{(T)}$ is the oscillatory coefficient defined in Eq. (22). Equivalently, it has the finite-mode form described in Eq. (23).

We then define the N -sample averaged extrapolated estimator

$$\hat{\theta}_{B,R}^{(N,m)} := \frac{1}{N} \sum_{j=1}^N \tilde{\theta}_{B,R}^{(m)}(T_j).$$

For a fixed realization $T_1, \dots, T_N$, averaging Eq. (33) over j gives

$$\begin{aligned} \hat{\theta}_{B,R}^{(N,m)} - \theta_B &= \frac{1}{N} \sum_{j=1}^N \sum_{k=2}^K \frac{\Phi_{k,m}^{(T_j)}}{T_j^k} \\ &\quad + O_{\text{nonosc}}(T^{-2m-2}) + O(T^{-K-1}). \end{aligned} \quad (34)$$

A. Analysis of the Bias

We now analyze the suppression of the oscillatory contribution under runtime randomization. Since the Richardson-extrapolated error contains all oscillatory orders

$$\sum_{k=2}^K \frac{\Phi_{k,m}^{(T_j)}}{T_j^k},$$

we introduce, for each k , the weighted characteristic function

$$\chi_{\mu,k}(\xi) := \int_{1-\lambda}^{1+\lambda} x^{-k} e^{i\xi x} d\mu(x). \quad (35)$$

The weight x^{-k} comes from the identity

$$T_j^{-k} = T^{-k} X_j^{-k}.$$

Using the finite-mode form of $\Phi_{k,m}^{(T)}$, we write

$$\Phi_{k,m}^{(T)} = \sum_{\nu \in I_k} [a_{k,\nu} \cos(\Omega_{k,\nu} T) + b_{k,\nu} \sin(\Omega_{k,\nu} T)],$$

where I_k is finite and $\Omega_{k,\nu} > 0$. Hence, for $T_j = T X_j$, we have

$$\frac{\Phi_{k,m}^{(T_j)}}{T_j^k} = \frac{1}{T^k} X_j^{-k} \Phi_{k,m}^{(T X_j)}.$$

Using the finite-mode expansion of $\Phi_{k,m}^{(T)}$, this gives

$$\mathbb{E} \left[\frac{\Phi_{k,m}^{(T_j)}}{T_j^k} \right] = \frac{1}{T^k} \sum_{\nu \in I_k} \left[a_{k,\nu} \int_{1-\lambda}^{1+\lambda} x^{-k} \cos(\Omega_{k,\nu} T x) d\mu(x) + b_{k,\nu} \int_{1-\lambda}^{1+\lambda} x^{-k} \sin(\Omega_{k,\nu} T x) d\mu(x) \right]. \quad (36)$$

By the definition of $\chi_{\mu,k}$, this implies

$$\left| \mathbb{E} \left[\frac{\Phi_{k,m}^{(T_j)}}{T_j^k} \right] \right| \leq \frac{1}{T^k} \sum_{\nu \in I_k} (|a_{k,\nu}| + |b_{k,\nu}|) |\chi_{\mu,k}(\Omega_{k,\nu} T)|. \quad (37)$$

Thus the oscillatory bias is governed by the decay of $\chi_{\mu,k}$ at the frequencies $\Omega_{k,\nu} T$.

Assume that, for some $M > 0$, the weighted characteristic functions satisfy

$$|\chi_{\mu,k}(\xi)| \leq C_{\mu,K} (1 + |\xi|)^{-M}, \quad 2 \leq k \leq K. \quad (38)$$

Since the frequencies are bounded away from zero, namely $\Omega_{k,\nu} \geq \Delta_{\min}$, we obtain

$$\begin{aligned} \left| \mathbb{E} \left[\sum_{k=2}^K \frac{\Phi_{k,m}^{(T_j)}}{T_j^k} \right] \right| &= O \left(\sum_{k=2}^K \frac{1}{T^k} \frac{1}{(\Delta_{\min} T)^M} \right) \\ &= O \left(\frac{1}{\Delta_{\min}^M T^{M+2}} \right). \end{aligned} \quad (39)$$

Thus the leading randomized oscillatory bias comes from the $k = 2$ sector, while all higher oscillatory orders are smaller.

Combining this with the non-oscillatory remainder after m -fold Richardson extrapolation gives

$$\begin{aligned} \mathbb{E}[\hat{\theta}_{B,R}^{(N,m)} - \theta_B] &= O \left(\frac{1}{\Delta_{\min}^M T^{M+2}} \right) \\ &\quad + O(T^{-2m-2}) + O(T^{-K-1}). \end{aligned} \quad (40)$$

The leading randomized oscillatory bias comes from the $k = 2$ sector. For this sector, the boundary estimate gives

$$\begin{aligned} \sum_{\nu \in I_2} (|a_{2,\nu}| + |b_{2,\nu}|) &= L_m(\alpha) \sum_{n \neq 0} |B_n| \\ &\leq L_m(\alpha) \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4}, \end{aligned} \quad (41)$$

where

$$L_m(\alpha) := \sum_{\ell=0}^m |w_{m,\ell}| \alpha^{-2\ell}$$

is the Richardson amplification factor for the leading T^{-2} sector. Therefore,

$$\left| \mathbb{E} \left[\frac{\Phi_{2,m}^{(T_j)}}{T_j^2} \right] \right| \leq C_{\mu,M} L_m(\alpha) \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 \Delta_{\min}^M T^{M+2}}. \quad (42)$$

Two representative choices of μ are worth noting.

a. Example 1: Uniform randomization. For asymptotic performance estimates, one can use a sharper large- T form. A single integration by parts gives

$$\begin{aligned} \chi_{\mu,k}(\xi) &= \frac{1}{2\lambda i \xi} \left[(1+\lambda)^{-k} e^{i\xi(1+\lambda)} - (1-\lambda)^{-k} e^{i\xi(1-\lambda)} \right] \\ &\quad + O(|\xi|^{-2}). \end{aligned}$$

Thus, when $|\xi| \gg 1$, the weighted characteristic function satisfies

$$|\chi_{\mu,k}(\xi)| \leq \frac{C_{\mu,k}}{|\xi|} + O(|\xi|^{-2}),$$

where the leading endpoint constant is

$$C_{\mu,k} = \frac{(1-\lambda)^{-k} + (1+\lambda)^{-k}}{2\lambda}.$$

Since $\xi = \Omega_{2,\nu} T$ and $|\Omega_{2,\nu}| \geq \Delta_{\min}$, this large- T estimate applies when $\Delta_{\min} T \gg 1$. In this regime, the leading $k = 2$ oscillatory bias is governed by

$$C_{\mu,2} = \frac{(1-\lambda)^{-2} + (1+\lambda)^{-2}}{2\lambda},$$

instead of the worst-case constant $(1-\lambda)^{-2}/\lambda$. For $\lambda = 0.7$, we obtain

$$C_{\mu,2} = \frac{(0.3)^{-2} + (1.7)^{-2}}{2(0.7)} \simeq 8.18.$$

b. Example 2: Triangular randomization. For triangular randomization, take

$$\frac{d\mu}{dx} = \frac{1}{\lambda} \left(1 - \frac{|x-1|}{\lambda} \right), \quad x \in [1-\lambda, 1+\lambda].$$

Then

$$\chi_{\mu,k}(\xi) = \int_{1-\lambda}^{1+\lambda} x^{-k} \frac{1}{\lambda} \left(1 - \frac{|x-1|}{\lambda} \right) e^{i\xi x} dx.$$

The density vanishes at the endpoints, so the $O(|\xi|^{-1})$ endpoint term is absent. Integrating by parts on $[1-\lambda, 1]$ and $[1, 1+\lambda]$ gives

$$\begin{aligned} \chi_{\mu,k}(\xi) &= \frac{(1-\lambda)^{-k} e^{i\xi(1-\lambda)}}{(i\xi)^2 \lambda^2} - \frac{2e^{i\xi}}{(i\xi)^2 \lambda^2} \\ &\quad + \frac{(1+\lambda)^{-k} e^{i\xi(1+\lambda)}}{(i\xi)^2 \lambda^2} + O(|\xi|^{-3}). \end{aligned}$$

Thus, when $|\xi| \gg 1$, the weighted characteristic function satisfies

$$|\chi_{\mu,k}(\xi)| \leq \frac{C_{\mu,k}}{|\xi|^2} + O(|\xi|^{-3}),$$

where the leading endpoint constant is

$$C_{\mu,k} = \frac{(1-\lambda)^{-k} + 2 + (1+\lambda)^{-k}}{\lambda^2}.$$

Thus triangular randomization corresponds to $M = 2$. Since $\xi = \Omega_{2,\nu}T$ and $|\Omega_{2,\nu}| \geq \Delta_{\min}$, this large- T estimate applies when $\Delta_{\min}T \gg 1$. In this regime, the leading $k = 2$ oscillatory bias is governed by

$$C_{\mu,2} = \frac{(1-\lambda)^{-2} + 2 + (1+\lambda)^{-2}}{\lambda^2}.$$

For $\lambda = 0.7$, we obtain

$$C_{\mu,2} = \frac{(0.3)^{-2} + 2 + (1.7)^{-2}}{(0.7)^2} \simeq 27.46.$$

c. Example 3: Smooth bump randomization. Finally, consider the normalized bump density

$$\frac{d\mu}{dx} = \frac{1}{Z_\lambda} \exp\left(-\frac{1}{1 - \left(\frac{x-1}{\lambda}\right)^2}\right), \quad x \in (1-\lambda, 1+\lambda),$$

and $d\mu/dx = 0$ outside this interval. Here Z_λ is the normalization constant. Since this density vanishes to all orders at the endpoints, $x^{-k}d\mu/dx$ is smooth and compactly supported for every fixed k . Repeated integration by parts gives, for every integer $M \geq 1$,

$$|\chi_{\mu,k}(\xi)| \leq \frac{C_{\mu,k}^{(M)}}{|\xi|^M},$$

where

$$C_{\mu,k}^{(M)} = \int_{1-\lambda}^{1+\lambda} \left| \frac{d^M}{dx^M} \left(x^{-k} \frac{d\mu}{dx} \right) \right| dx.$$

Thus smooth bump randomization gives arbitrary inverse-power suppression, with an M -dependent constant.

For the leading $k = 2$ sector, the relevant constant is

$$C_{\mu,2}^{(M)} = \int_{1-\lambda}^{1+\lambda} \left| \frac{d^M}{dx^M} \left(x^{-2} \frac{d\mu}{dx} \right) \right| dx.$$

For the numerical choice $\lambda = 0.7$, the standard bump gives approximately

$$C_{\mu,2}^{(1)} \simeq 4.06, \quad C_{\mu,2}^{(2)} \simeq 34.34.$$

Consequently, for $M = 2$, the leading randomized oscillatory bias is bounded by

$$\left| \mathbb{E} \left[\frac{\Phi_{2,m}^{(T_j)}}{T_j^2} \right] \right| \leq C_{\mu,2}^{(2)} L_m(\alpha) \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 \Delta_{\min}^2 T^4}.$$

Using $C_{\mu,2}^{(2)} \simeq 34.34$, this becomes

$$\left| \mathbb{E} \left[\frac{\Phi_{2,m}^{(T_j)}}{T_j^2} \right] \right| \leq 34.34 L_m(\alpha) \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 \Delta_{\min}^2 T^4}.$$

More generally, by increasing M , the oscillatory bias can be suppressed to arbitrary inverse order in T , up to the

remaining non-oscillatory Richardson error $O(T^{-2m-2})$ and the truncation remainder $O(T^{-K-1})$. More generally, by increasing M , the oscillatory bias can be suppressed to arbitrary inverse order in T , up to the remaining non-oscillatory Richardson error $O(T^{-2m-2})$ and the truncation remainder $O(T^{-K-1})$. For a fixed runtime, however, increasing M does not necessarily give a tighter numerical bound, because the integration-by-parts constant $C_{\mu,K,M}$ also grows with M . Thus the sharpest finite- T bound is obtained by minimizing $C_{\mu,K,M}(\Delta_{\min}T)^{-M}$ over the available choices of M . In the asymptotic regime $\Delta_{\min}T \gg 1$, higher values of M become advantageous and reflect the super-polynomial suppression provided by the smooth bump.

The random runtime $T_j = TX_j$ is easy to sample classically; no change to the adiabatic protocol itself is required. The smooth bump is therefore the best asymptotic choice among the elementary distributions considered here. For moderate runtimes, however, the larger constants in the bump bounds can make uniform or triangular randomization competitive, or even better at selected frequencies.

B. Analysis of the Variance

We next estimate the finite-sample statistical error of the randomized protocol. The random oscillatory part of the m -fold Richardson estimator is

$$Y(T_j) := \sum_{k=2}^K \frac{\Phi_{k,m}^{(T_j)}}{T_j^k}, \quad T_j = TX_j. \quad (43)$$

Here X_j are drawn independently from μ , supported on $[1-\lambda, 1+\lambda]$ with $0 < \lambda < 1$. Define the sample average

$$\bar{Y}_N := \frac{1}{N} \sum_{j=1}^N Y(T_j). \quad (44)$$

Since the samples are independent,

$$\text{Var}(\bar{Y}_N) = \frac{1}{N} \text{Var}(Y(T_j)).$$

We now bound the variance of a single sample. Since $X_j \in [1-\lambda, 1+\lambda]$, we have

$$T_j \in [(1-\lambda)T, (1+\lambda)T],$$

and therefore

$$T_j^{-k} \leq (1-\lambda)^{-k} T^{-k}.$$

Using the finite-mode form of $\Phi_{k,m}^{(T)}$, define

$$A_k := \sum_{\nu \in I_k} (|a_{k,\nu}| + |b_{k,\nu}|).$$

Then

$$|Y(T_j)| \leq \sum_{k=2}^K \frac{A_k}{(1-\lambda)^k T^k} = O(A_2 T^{-2}) + O(T^{-3}), \quad (45)$$

where K is fixed and the constants may depend on λ, K, α . Hence

$$\text{Var}(Y(T_j)) \leq \mathbb{E}[|Y(T_j)|^2] = O(A_2^2 T^{-4}) + O(T^{-5}). \quad (46)$$

Consequently,

$$\sqrt{\text{Var}(\bar{Y}_N)} = O\left(\frac{A_2}{T^2 \sqrt{N}}\right) + O\left(\frac{1}{T^3 \sqrt{N}}\right). \quad (47)$$

For the leading $k = 2$ sector, the boundary estimate gives

$$A_2 = O\left(\frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4}\right).$$

Thus the finite-sample fluctuation satisfies

$$\sqrt{\text{Var}(\bar{Y}_N)} = O\left(\frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 T^2 \sqrt{N}}\right). \quad (48)$$

Equivalently, the statistical fluctuation is of this order in probability.

This shows that runtime randomization suppresses the oscillatory contribution at the level of the mean, through the decay of the weighted characteristic functions, but it does not generically improve the finite-sample fluctuation scaling. The latter remains of order $T^{-2}N^{-1/2}$, governed by the leading $k = 2$ oscillatory sector.

Under the additional endpoint condition $\dot{H}(0) = \dot{H}(1)$, the leading bound simplifies to

$$\sqrt{\text{Var}(\bar{Y}_N)} = O\left(\frac{\|\dot{H}(0)\|^2}{\Delta(0)^4 T^2 \sqrt{N}}\right). \quad (49)$$

C. Combined error

The preceding analysis shows that the deterministic and statistical contributions to the error behave differently under extrapolation and runtime randomization. Forward–reverse cancellation and Richardson extrapolation remove the non-oscillatory adiabatic terms up to the chosen order, while runtime randomization suppresses the remaining oscillatory terms in expectation.

For any fixed extrapolation order m , however, runtime randomization does not remove the finite-sample fluctuation of the estimator. The variance analysis gives

$$\sqrt{\text{Var}(\bar{Y}_N)} = O\left(\frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 T^2 \sqrt{N}}\right), \quad (50)$$

where $\bar{Y}_N$ denotes the sample average of the oscillatory part of the randomized estimator. Thus runtime randomization suppresses the oscillatory bias through destructive averaging, but the residual sample fluctuation remains of order $T^{-2}N^{-1/2}$.

Consequently, once the deterministic bias has been reduced below the scale in Eq. (50), the protocol is no longer bias-limited but statistics-limited. Further improvement then requires increasing the number of randomized samples N , rather than removing additional deterministic terms. In particular, taking $N = O(\epsilon^{-2})$ makes the statistical fluctuation compatible with a target additive precision ϵ .

Theorem 3 (Runtime-randomized Richardson estimator). *Under the setting of Section II, let $X_1, \dots, X_N$ be sampled independently from a probability distribution μ supported on $[1-\lambda, 1+\lambda]$, with $0 < \lambda < 1$, and set*

$$T_j := TX_j.$$

Let

$$\hat{\theta}_{B,R}^{(N,m)}(T) := \frac{1}{N} \sum_{j=1}^N \tilde{\theta}_{B,R}^{(m)}(T_j)$$

be the randomized m -fold Richardson estimator. Suppose that, for some $M > 0$, the weighted characteristic functions

$$\chi_{\mu,k}(\xi) := \int_{1-\lambda}^{1+\lambda} x^{-k} e^{i\xi x} d\mu(x)$$

satisfy

$$|\chi_{\mu,k}(\xi)| \leq C_{\mu,K} (1 + |\xi|)^{-M}, \quad 2 \leq k \leq K.$$

Then, for any fixed integer $K \geq 2m+2$, the bias satisfies

$$\mathbb{E}[\hat{\theta}_{B,R}^{(N,m)}(T)] - \theta_B = O\left(\frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 \Delta_{\min}^M T^{M+2}}\right) + O(T^{-2m-2}) + O(T^{-K-1}). \quad (51)$$

The first term is the randomized leading oscillatory bias, the second term is the remaining non-oscillatory Richardson error, and the last term is the truncation remainder. Moreover, the standard deviation of the randomized estimator satisfies

$$\sqrt{\text{Var}(\hat{\theta}_{B,R}^{(N,m)}(T))} \leq L_m(\alpha) \sqrt{\mathbb{E}[X^{-4}]} \frac{\|\dot{H}(0)\| \|\dot{H}(1)\|}{\Delta(0)^4 T^2 \sqrt{N}}.$$

The parameter M in Theorem 3 is the decay order of the weighted characteristic functions of the runtime-randomization distribution. For uniform randomization, one obtains $M = 1$, giving the oscillatory bias scaling $O(T^{-3})$. For a smooth compactly supported runtime distribution whose density vanishes to all orders at the endpoints, the bound on $\chi_{\mu,k}$ holds for every fixed M , with

a constant depending on M , K , and μ . Thus the oscillatory bias can be suppressed to arbitrary fixed inverse-polynomial order.

This does not by itself remove the non-oscillatory Richardson remainder $O(T^{-2m-2})$. To make the non-oscillatory remainder no larger than the randomized oscillatory bias $O(T^{-M-2})$, it is enough to choose $m \geq M/2$. Thus runtime randomization controls the oscillatory sector, while Richardson extrapolation controls the non-oscillatory sector.

V. NUMERICAL SIMULATIONS

We test the cancellation cascade of Section III on two models: an exactly solvable spin-1/2 system in a cone field and an entangled spiral Heisenberg chain. For each model, we first reconstruct the Berry phase at a small runtime and then compare the finite-time errors of four estimators: the single forward estimator, the forward–reverse estimator, and the Richardson-extrapolated estimators with one and two Richardson levels combined with smooth runtime randomization.

The simulations exhibit the effective scaling hierarchy

$$T^{-1} \xrightarrow{\text{fwd-rev}} T^{-2} \xrightarrow{\text{1-fold Richardson+bump}} T^{-4}$$

$$T^{-1} \xrightarrow{\text{fwd-rev}} T^{-2} \xrightarrow{\text{2-fold Richardson+bump}} T^{-6}.$$

Forward–reverse symmetrization cancels the odd non-oscillatory terms. The subsequent improvements come from the combined effect of Richardson extrapolation and runtime randomization: Richardson extrapolation removes successive even non-oscillatory terms, while runtime randomization suppresses the residual oscillatory contribution in expectation. Thus the combined protocol reveals the higher-order scaling in the numerical data.

A third set of simulations benchmarks the analytic bounds and identifies which spectral gap controls each stage. As a baseline, we compare with the runtime-scaling reconstruction of Ref. [1], which uses two forward evolutions at T and αT .

A. Spin-1/2 cone loop

As a controlled testbed, we sweep a spin-1/2 system in a field of fixed magnitude around a cone of half-angle θ_0 on the Bloch sphere,

$$H(s) = -\frac{1}{2} \mathbf{B}(s) \cdot \boldsymbol{\sigma}.$$

The ground-state Berry phase is known exactly,

$$\theta_B = -\pi(1 - \cos \theta_0).$$

Figure 1(a) reconstructs θ_B at the small runtime $T = 8$. The runtime-scaling method of Ref. [1] uses two forward evolutions at T and αT . Since the Berry phase

is runtime-independent, whereas the dynamical phase rescales with the runtime, the two eigenphase estimates can be combined to cancel the dynamical phase without using spectral information. At this short runtime, however, the remaining adiabatic error is still large, of order one radian, and the estimate visibly departs from the exact curve. By contrast, the two-fold Richardson + bump protocol agrees with the analytic value at the 10^{-2} – 10^{-3} radian level for the same runtime.

Figure 1(b) traces the error versus runtime at $\theta_0 = 0.4\pi$, where the gap is $\Delta = 1$. A single forward traversal has the expected $O(T^{-1})$ adiabatic phase error. The forward–reverse estimator cancels the odd non-oscillatory terms, producing the $O(T^{-2})$ scaling described in Theorem 1. Combining smooth runtime randomization with one-fold and two-fold Richardson extrapolation then gives the observed $O(T^{-4})$ and $O(T^{-6})$ scalings, respectively. In this combined protocol, Richardson extrapolation removes the relevant even non-oscillatory terms, while runtime randomization suppresses the residual oscillatory contribution in expectation, consistently with Theorems 2 and 3. The dips in the curves correspond to zero crossings of the signed residual bias.

B. Theory bounds and the controlling gap

To check the analytic bounds and identify which gap governs each stage, we deform the loop into a non-isospectral one by modulating the field magnitude,

$$|\mathbf{B}(s)| = |\mathbf{B}|(1 - a \sin^2(\pi s)), \quad a = 0.4.$$

This lowers the gap in the *interior* of the loop to $\Delta_{\min} = 0.6$, while leaving the *endpoint* quantities—the gap $\Delta(0) = 1$ and the rate $\|\dot{H}(0)\|$ —unchanged. Since the direction of the field is unchanged, the Berry phase θ_B remains invariant, as shown in Fig. 2(a).

The single-evolution error is set by the interior worst-case gap,

$$|\varphi| \lesssim \frac{\|\dot{H}\|_{\max}^2}{\Delta_{\min}^3 T},$$

so the dip raises both the numerics and, more steeply, the bound (Fig. 2(b)). The Richardson residual, by contrast, is an endpoint-controlled boundary term,

$$\frac{\|\dot{H}(0)\|^2}{\Delta(0)^4 T^2}.$$

Thus the dipped and undipped data fall on the *same* line, independent of the interior gap (Fig. 2(c)). Finally, the randomized one-fold Richardson bias obeys the estimate in Theorem 3. For a C^∞ bump with effective decay order $M = 2$, the bound reads

$$C \frac{\|\dot{H}(0)\|^2}{\Delta(0)^4 \Delta_{\min}^2 T^4}.$$

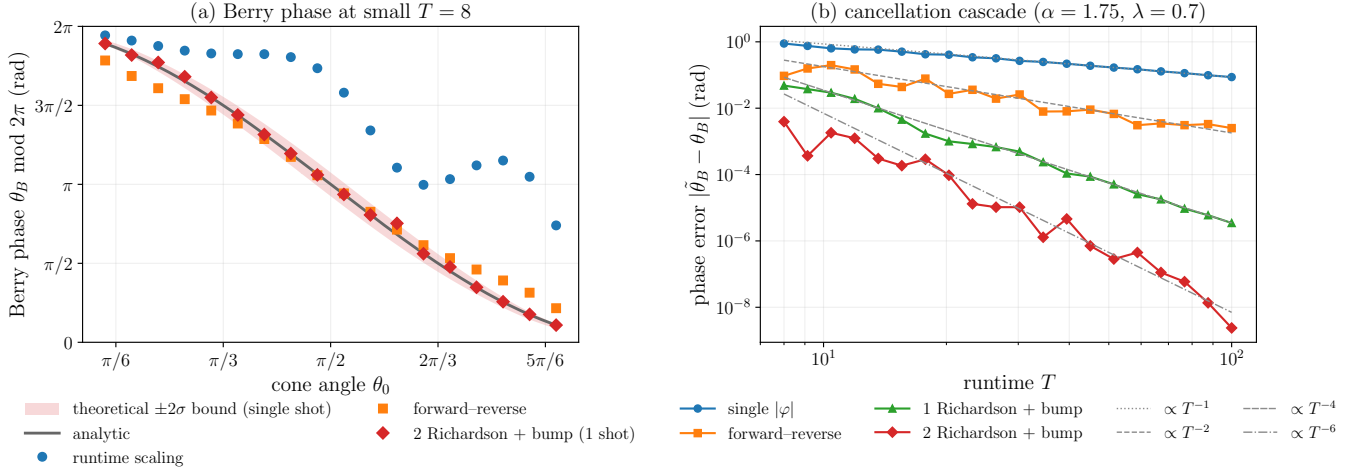


FIG. 1. Spin-1/2 cone loop. **(a)** Berry phase θ_B versus cone half-angle θ_0 at a fixed small runtime $T = 8$: the exact analytic value $-\pi(1 - \cos \theta_0)$ (line), the runtime-scaling reconstruction from two forward evolutions at T and αT ($\circ$; Ref. [1]), and the two-fold Richardson + bump estimate ($\diamond$). At this cheap runtime, the runtime-scaling estimate is visibly off the curve, with a residual adiabatic error of order one radian, while the proposed protocol lies on it. **(b)** Error-cancellation cascade versus runtime ($\theta_0 = 0.4\pi, \Delta = 1$): single evolution ($\circ$, $\propto T^{-1}$), forward-reverse ($\square$, $\propto T^{-2}$), one-fold Richardson + bump randomization ($\triangle$, $\propto T^{-4}$), and two-fold Richardson + bump randomization ($\diamond$, $\propto T^{-6}$). Grey lines are reference slopes, with $\lambda = 0.7$ and $\alpha = 1.75$. The two-fold curve sits at its T^{-6} floor and is oscillation-limited.

A single constant $C \approx 48$ upper-bounds both curves (Fig. 2(d)), confirming the joint dependence on the endpoint gap $\Delta(0)$ and the interior gap $\Delta_{\min}$.

C. Entangled spiral Heisenberg chain

We repeat the programme on a genuinely interacting model: a four-site Heisenberg chain

$$H(s) = - \sum_i \mathbf{B}_i(s) \cdot \mathbf{S}_i + J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad J = 1,$$

with a site-dependent azimuthal offset $\phi_i = 2\pi i/N$ (a “spiral”), which breaks the global spin-rotation symmetry so that the ground state is entangled and the virtual-excitation contribution $A^{(2)}$ to the phase error is nonzero — a true many-body test. The rigid global rotation underlying the loop yields an exact propagator and a closed-form reference $\theta_B = 2\pi \langle S_{\text{tot}}^z \rangle_0$.

Figure 3(a) plots θ_B versus cone angle at a small runtime $T = 20$ (the gap varies over $[0.11, 0.42]$ across the sweep): the runtime-scaling estimate departs visibly from the analytic $2\pi \langle S_{\text{tot}}^z \rangle_0$ while the protocol’s estimate lies on it. Figure 3(b) applies the same four estimators unchanged at $\theta = 0.4\pi$ (gap = 0.37) and reproduces the full hierarchy $O(T^{-1}) \rightarrow O(T^{-2}) \rightarrow O(T^{-4}) \rightarrow O(T^{-6})$. The smaller gap enlarges the prefactors, so we take $T \in [20, 200]$ and a wider randomization window $\lambda = 0.8$ ($\lambda \Delta_{\min} T \gtrsim 1$). The cancellation thus persists for a genuinely entangled many-qubit adiabatic loop.

VI. BERRY PHASE ESTIMATION ALGORITHM

We now describe the algorithmic implementation of the cancellation and extrapolation results developed above. The main protocol is formulated in terms of Hadamard-test estimation of overlap amplitudes. This choice is natural in the present setting, since the relevant phase information is contained in the survival amplitude

$$z(T) = \langle \psi(0) | U_T(1) | \psi(0) \rangle,$$

whose real and imaginary parts can be estimated directly at each randomized runtime. Although this shot-based approach has sample complexity $O(\varepsilon^{-2})$ for estimating an amplitude to accuracy ε , it allows the statistical error from runtime randomization to be treated separately from the deterministic adiabatic bias. Moreover, leakage only reduces the overlap signal multiplicatively, and therefore does not change the asymptotic runtime scaling.

A QPE-based implementation is also possible and is described in Appendix E. While QPE is efficient for resolving a fixed eigenphase, it is less natural for the randomized estimators considered here, since runtime randomization must be combined with controlled powers $U_T(1)^{2^k}$ and leakage becomes a sampling failure. We therefore use Hadamard tests as the main primitive for the high-order randomized algorithm.

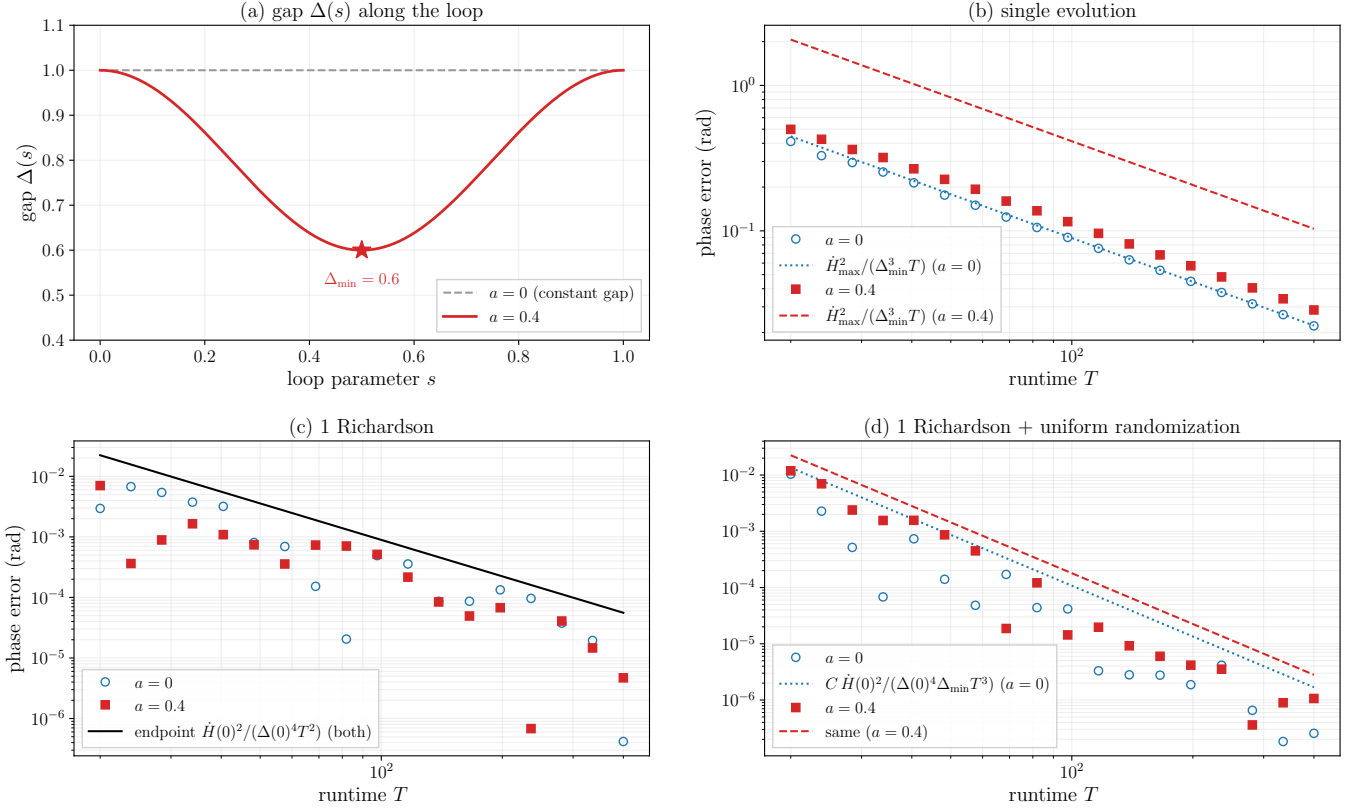


FIG. 2. Numerics against the analytic bounds, and the controlling gap for each estimator, on a non-isospectral spin-1/2 loop with field modulation $|\mathbf{B}(s)| = |\mathbf{B}|(1 - a \sin^2(\pi s))$, $a = 0.4$. **(a)** Instantaneous gap $\Delta(s)$: constant case ($a = 0$, dashed) versus a loop dipped to $\Delta_{\min} = 0.6$ in the interior (solid), with the endpoints fixed at $\Delta(0) = 1$. **(b)** Single-evolution error with the worst-case bound $\|\dot{H}\|_{\max}^2/(\Delta_{\min}^3 T)$ for the dipped and undipped loops: the error tracks the *interior* gap $\Delta_{\min}$. **(c)** One-fold Richardson residual: dipped and undipped data collapse onto the same endpoint-controlled line $\|\dot{H}(0)\|^2/(\Delta(0)^4 T^2)$, independent of $\Delta_{\min}$. **(d)** One-fold Richardson with bump randomization and the Theorem 3 bound with effective decay order $M = 2$, $C\|\dot{H}(0)\|^2/(\Delta(0)^4 \Delta_{\min}^2 T^4)$; one constant $C \approx 48$ bounds both loops. Here $T \in [20, 400]$.

A. Hadamard-test based algorithm

Let $r \geq 0$ be the Richardson level, so that the target deterministic bias order is T^{-2r-2} . Let $\alpha > 1$, and choose Richardson weights $w_{r,\ell}$, $\ell = 0, \dots, r$, satisfying

$$\sum_{\ell=0}^r w_{r,\ell} = 1, \quad \sum_{\ell=0}^r w_{r,\ell} \alpha^{-2q\ell} = 0, \quad q = 1, \dots, r.$$

Let ρ be a probability density supported on $[1 - \lambda, 1 + \lambda]$, with $\lambda \in (0, 1)$, whose characteristic function satisfies

$$|\hat{\rho}(\xi)| \leq C_{\rho,r} (1 + |\xi|)^{-2r-2}.$$

A compactly supported C^∞ bump density is one convenient choice.

Step 1: Coarse branch resolution. We first obtain a coarse estimate $\hat{\theta}_B^{\text{coarse}}$ of θ_B over the full range $[0, 2\pi)$, sufficient to identify the correct π -branch in the subsequent fine estimates. Since only constant accuracy is required, this step can be implemented using Hadamard

tests at appropriately chosen runtimes. Its cost is independent of the target fine accuracy ε_B , and is therefore subdominant in the overall complexity. The concrete construction of this coarse branch estimate is described in the next subsection.

Step 2: Runtime randomization. For each $j = 1, \dots, N$, sample

$$X_j \sim \rho,$$

and define the Richardson-node runtimes

$$T_{j,\ell} := \alpha^\ell T X_j, \quad \ell = 0, \dots, r.$$

Step 3: Hadamard-test sampling. For each pair (j, ℓ) , perform Hadamard tests for the forward and reverse propagators

$$U_{T_{j,\ell}}(1), \quad \hat{U}_{T_{j,\ell}}(1).$$

Using the appropriate control phases, estimate the real and imaginary parts of the corresponding overlap am-

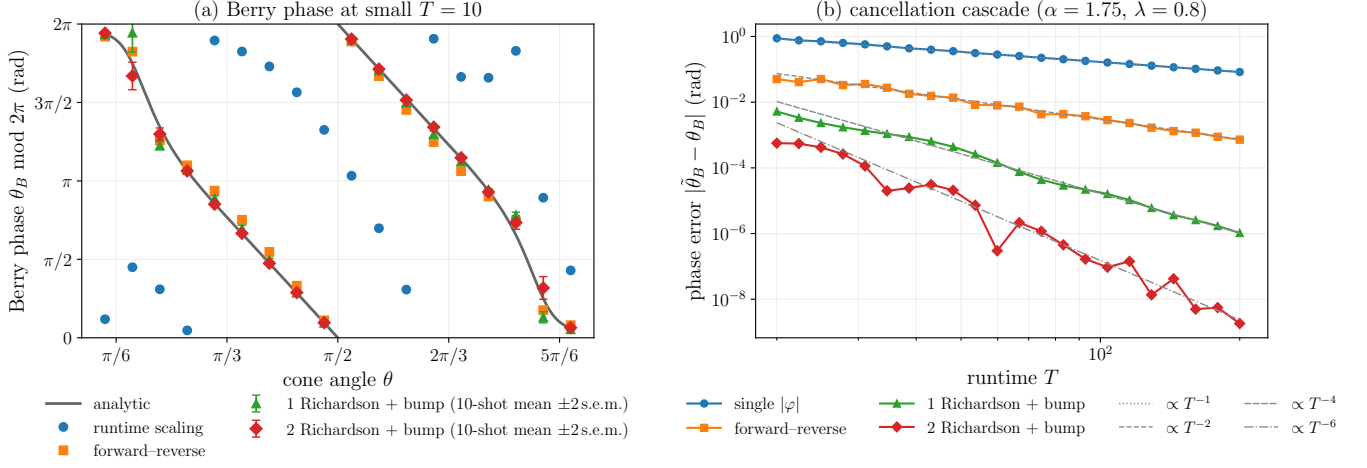


FIG. 3. Entangled spiral Heisenberg chain ($N = 4$, $J = 1$, $B_0 = 1$). **(a)** Berry phase θ_B versus cone angle θ at a fixed small runtime $T = 20$: the exact analytic value $2\pi\langle S_{\text{tot}}^z \rangle_0$ (line), the runtime-scaling reconstruction from two forward evolutions at T and αT ($\circ$; Ref. [1]), and the two-Richardson + bump estimate ($\diamond$); the runtime-scaling estimate departs from the curve while the protocol lies on it (the gap over the sweep lies in $[0.11, 0.42]$). **(b)** Error-cancellation cascade versus runtime ($\theta = 0.4\pi$, gap = 0.37, $\theta_B = 1.4530$) with the same four estimators as in Fig. 1(b) ($\circ T^{-1}$, $\square T^{-2}$, $\triangle T^{-4}$, $\diamond T^{-6}$; grey reference slopes), $\alpha = 1.75$, $\lambda = 0.8$. The smaller gap pushes the clean T^{-6} regime to larger T .

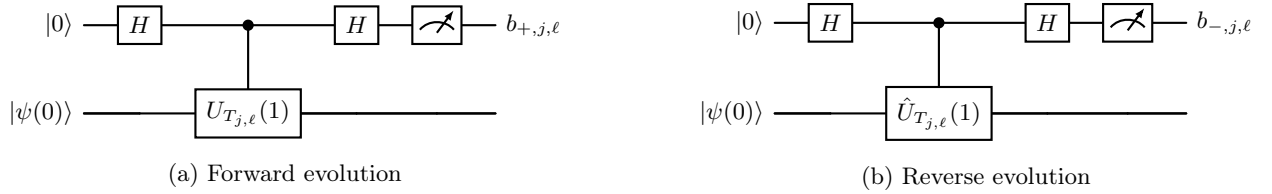


FIG. 4. Randomized Hadamard-test subroutine used in the Berry phase estimation algorithm. For each sample j and Richardson node ℓ , the runtime $T_{j,\ell} = \alpha^\ell T X_j$ is used in the forward and reverse adiabatic evolutions. Hadamard tests estimate the corresponding overlap amplitudes, from which a mod- π forward–reverse Berry phase estimate is reconstructed.

plitudes. From these estimates reconstruct a forward–reverse Berry phase estimate

$$\tilde{\theta}_B(T_{j,\ell}) \pmod{\pi}.$$

Step 4: Branch lifting. Using the coarse estimate $\hat{\theta}_B^{\text{coarse}}$, lift each mod- π estimate to the unique representative in the interval

$$I := (\hat{\theta}_B^{\text{coarse}} - \pi/2, \hat{\theta}_B^{\text{coarse}} + \pi/2).$$

We denote the lifted value by

$$\theta_B^{\text{lift}}(T_{j,\ell}).$$

Step 5: Richardson extrapolation and averaging.

For each sample j , form the Richardson combination

$$\Theta_{R,j}(T) := \sum_{\ell=0}^r w_{r,\ell} \theta_B^{\text{lift}}(T_{j,\ell}).$$

Finally output

$$\hat{\theta} := \left(\frac{1}{N} \sum_{j=1}^N \Theta_{R,j}(T) \right) \pmod{2\pi}.$$

Error and cost. Conditioned on successful branch lifting, the deterministic bias is $O(T^{-2r-2})$. Using the endpoint-bound prefactor in the complexity table, we use the asymptotic runtime scaling

$$T = O\left(\frac{(\|\dot{H}(0)\| \|\dot{H}(1)\|)^{1/(2r+2)}}{\Delta(0)^{2/(r+1)} \Delta_{\min}^{r/(r+1)} \varepsilon_B^{1/(2r+2)}} \right).$$

The statistical fluctuation consists of two contributions: the Hadamard-test shot noise and the fluctuation from sampling randomized runtimes. The shot-noise contribution is $O_p(N^{-1/2})$, so we choose

$$N = O\left(\varepsilon_B^{-2} \log \frac{1}{\eta} \right)$$

to make it $O(\varepsilon_B)$ with high probability, up to a constant redistribution of the failure budget. With this choice of N , the runtime-randomization fluctuation is subleading. Indeed, by Theorem 3, it scales as

$$O_p\left(\frac{1}{T^2 \sqrt{N}} \right),$$

up to path-dependent constants. Hence it is smaller than the shot-noise term by a factor T^{-2} in the adiabatic regime.

Finally, leakage does not change the deterministic cancellation mechanism. In the Hadamard-test implementation, leakage reduces the overlap magnitude and hence multiplies the phase-estimation noise by

$$(1 - p_{\text{leak}}^{\max})^{-1/2},$$

where $p_{\text{leak}}^{\max}$ is the maximum leakage probability over the runtimes used by the algorithm. Since

$$p_{\text{leak}}^{\max} = O(T^{-2}),$$

we have

$$(1 - p_{\text{leak}}^{\max})^{-1/2} = 1 + O(T^{-2}).$$

Thus leakage changes only the constants, not the asymptotic ε_B -scaling, provided the algorithm is run in the adiabatic regime where $p_{\text{leak}}^{\max} \ll 1$.

B. Branch lifting

The forward–reverse estimator in Eq. (11) determines θ_B only modulo π . Before applying Richardson extrapolation or runtime randomization, we therefore need a coarse mod- 2π estimate to choose the correct π -branch. This preprocessing step only requires constant accuracy; the final high-precision scaling is supplied by the branch-lifted forward–reverse Richardson and randomized estimator.

The same branch-resolution step can also be viewed as a binary discriminator in symmetry-quantized settings. For example, the Zak phase of a one-dimensional inversion-symmetric band is restricted to 0 or π modulo 2π . We do not pursue this application here.

Concretely, suppose that a coarse estimate θ_{br} satisfies

$$|\theta_{\text{br}} - \theta_B| < \eta_c \pmod{2\pi}$$

and that each fine forward–reverse estimate satisfies

$$|\tilde{\theta}_B(T_\ell) - \theta_B| < \eta_f \pmod{\pi}.$$

If

$$\eta_c + \eta_f < \frac{\pi}{2},$$

then each fine estimate has a unique representative in

$$I_{\text{br}} = \left(\theta_{\text{br}} - \frac{\pi}{2}, \theta_{\text{br}} + \frac{\pi}{2} \right),$$

and this representative lies on the correct branch. After this lifting step, Richardson extrapolation and runtime randomization can be applied as ordinary real linear operations.

We now construct such a coarse estimate. We use the runtime-scaling branch-resolution framework of Ref. [1]. In the nearby-runtime version, the relative runtime is chosen close to one using a Hamiltonian-norm bound, so that the branch information is extracted from a small phase difference. Here we instead use fixed integer runtime ratios. This avoids a near-degenerate phase comparison depending on T and $H_{\max}$, and the same integer conditions also cancel the leading non-oscillatory adiabatic error.

Let

$$\vartheta_+(T) := -\theta_D(T) + \theta_B + \varphi(T) \pmod{2\pi}$$

be the forward eigenphase associated with the transported ground state. Here $\theta_D(T)$ is exactly linear in T , while by Lemma 1,

$$\varphi(T) = \frac{\varphi_1}{T} + \sum_{k=2}^K \frac{\varphi_k + \varphi_k^{(T)}}{T^k} + O(T^{-K-1}).$$

Choose positive runtime ratios $q_1, \dots, q_m$ and integer coefficients $n_1, \dots, n_m$, and define

$$\Theta_{\mathbf{n}}(T) := \sum_{j=1}^m n_j \vartheta_+(q_j T) \pmod{2\pi}.$$

Because the coefficients are integers, $\Theta_{\mathbf{n}}(T)$ is well defined modulo 2π without unwrapping the individual eigenphases. We impose

$$\sum_{j=1}^m n_j = 1, \tag{52}$$

$$\sum_{j=1}^m n_j q_j = 0, \tag{53}$$

$$\sum_{j=1}^m n_j q_j^{-r} = 0, \quad r = 1, \dots, p. \tag{54}$$

These conditions keep one copy of θ_B , cancel the dynamical phase, and cancel the non-oscillatory adiabatic terms $T^{-1}, \dots, T^{-p}$. Substitution gives

$$\begin{aligned} \Theta_{\mathbf{n}}(T) &= \theta_B + \sum_{k=p+1}^K \frac{\varphi_k}{T^k} \sum_{j=1}^m n_j q_j^{-k} \\ &\quad + \sum_{k=2}^K \frac{1}{T^k} \sum_{j=1}^m n_j q_j^{-k} \varphi_k^{(q_j T)} + O(T^{-K-1}) \pmod{2\pi}. \end{aligned} \tag{55}$$

Thus the non-oscillatory residual starts at order T^{-p-1} . The oscillatory boundary terms generally remain, but branch lifting only needs constant accuracy.

As a simple low-order instance, take

$$(q_1, q_2, q_3) = (1, 2, 4), \quad (n_1, n_2, n_3) = (-2, 5, -2).$$

Then the above conditions hold with $p = 1$. Hence

$$\Theta_I(T) := -2\vartheta_+(T) + 5\vartheta_+(2T) - 2\vartheta_+(4T) \pmod{2\pi}$$

satisfies

$$\Theta_I(T) = \theta_B + O(T^{-2}) \pmod{2\pi}.$$

Thus the integer combination improves the asymptotic order of the coarse branch estimate from T^{-1} to T^{-2} .

This improvement comes with a constant overhead. One evaluation of $\Theta_I(T)$ uses the three runtimes $T, 2T, 4T$, whose total evolution time is $7T$. In addition, the integer coefficients amplify statistical errors: for independent phase estimates with comparable variance, the standard deviation is multiplied by

$$\sqrt{(-2)^2 + 5^2 + (-2)^2} = \sqrt{33},$$

while a deterministic worst-case error bound is multiplied by

$$|-2| + |5| + |-2| = 9.$$

Since branch lifting only requires constant accuracy and is performed at a coarse preprocessing runtime, this constant overhead may dominate the asymptotic T^{-2} improvement in finite- T regimes. This improvement should be interpreted with care: branch lifting only requires constant accuracy and is used at a relatively small preprocessing runtime, where the extra runtimes and larger integer coefficients may dominate the gain.

In this work we use the construction only at the level of asymptotic branch selection. A certified finite-precision choice of T_{br} would require a non-asymptotic adiabatic bound, including phase error and leakage, as in Ref. [35]. We leave this finite-time certification to future work.

Remark 1 (Branch consistency). Branch lifting assumes that all fine estimates used in the same Richardson or randomized linear combination are lifted to the same local branch. If the runtime variation is too large, or if the fine-estimator error is not uniformly controlled over the runtimes used, this branch consistency can fail.

In this work we do not give a rigorous finite-time safe bound guaranteeing branch consistency. Such a bound would require explicit non-asymptotic control of the fine-estimator error over the full runtime support, including phase error, leakage, and sampling error. We instead focus on the asymptotic cancellation order and the resulting precision scaling.

VII. DISCUSSION AND CONCLUSION

We have analyzed the adiabatic phase error in Berry phase estimation and established several results that improve both the theoretical understanding and the algorithmic efficiency of this problem.

Our main technical contribution is Lemma 1, which gives an explicit expansion of the phase error through $O(T^{-2})$ in terms of the Berry connections M_{nm} , spectral gaps Δ_{n0} , and the second-order adiabatic connection $\mathcal{A}^{(2)}$. Building on this, Theorem 1 shows that when the forward evolution under H and the reverse evolution under $-H$ are combined, not only the leading $O(T^{-1})$ phase error but, more generally, all odd-order non-oscillatory contributions cancel. As a result, the net systematic error starts at $O(T^{-2})$. This cancellation is deterministic and universal: it holds for any gapped Hamiltonian family satisfying the loop conditions, without requiring time-reversal symmetry or any other structural assumption beyond smoothness.

This error cancellation has direct algorithmic consequences. By running quantum phase estimation separately on $U_T(1)$ and $\hat{U}_T(1)$ and combining the results classically, the required adiabatic runtime is reduced from $T = O(1/\varepsilon_B)$ to $T = O(1/\sqrt{\varepsilon_B})$. We further showed that Richardson extrapolation eliminates the non-oscillatory part of the residual $O(T^{-2})$ term, so that the leading remaining error is no longer governed by bulk information along the path, but only by endpoint data. More precisely, its coefficient is controlled by the endpoint ground-to-excited couplings, scaling as $O(\|\hat{H}(0)\|^2/\Delta(0)^4)$. The remaining $O(T^{-2})$ contribution is purely oscillatory and therefore cannot be removed by deterministic extrapolation alone.

To address this residual oscillatory error, we introduced uniform runtime randomization in a Hadamard-test implementation. We showed that the leading oscillatory $O(T^{-2})$ contribution is suppressed by one additional factor of $1/T$ in expectation, so that the resulting deterministic bias becomes $O(T^{-3})$. Combined with the standard $O(N^{-1/2})$ shot-noise scaling of the Hadamard test, this yields a randomized algorithm for Berry phase estimation over the full phase range $[0, 2\pi)$ with runtime scaling $O(\varepsilon_B^{-1/3})$ under standard sample complexity.

Taken together, our results identify Berry phase estimation as a particularly promising target for near-term quantum computation. Unlike standard energy-estimation problems, it possesses a natural and intrinsic adiabatic error-cancellation mechanism rooted in the geometric structure of the quantity itself, and this structure can be further exploited by simple classical post-processing and runtime randomization. This suggests that Berry phases, and possibly geometric quantities more generally, may form a class of problems that can be estimated with high accuracy even when coherent resources are limited. In particular, in the early fault-tolerant regime, the large shot budgets available in Hadamard-test-based algorithms are not merely a cost, but a natural resource for suppressing adiabatic error. More broadly, our work points to the possibility that a deeper understanding of quantum algorithms for geometric quantities could lead to a broader theoretical framework for practical quantum advantage in early fault-tolerant quantum computation.

A. Robustness Against Other Sources of Error

In this work, we have focused on adiabatic error arising from finite-time evolution. For practical implementations, however, other sources of error are also important, including discretization error in digital simulation and circuit noise from imperfect gates and measurements.

It is natural to ask whether the robustness mechanism identified here extends beyond adiabatic error. Since the Berry phase is a geometric quantity, one may expect a certain degree of robustness also against local implementation errors. In particular, the forward–reverse structure of our protocol may help mitigate some errors that affect the two branches in a correlated way.

A full analysis of discretization error and circuit noise is beyond the scope of the present work. Clarifying whether geometric quantities such as the Berry phase enjoy a broader form of robustness remains an interesting direction for future work.

B. Generalization to Chern numbers and non-abelian holonomies

The Berry phase considered in this work is a $U(1)$ holonomy associated with a non-degenerate ground state transported around a single loop in parameter space. Two natural generalizations arise.

First, the Chern number—a topological invariant defined as the integral of the Berry curvature over a two-dimensional parameter surface—can in principle be computed by decomposing the surface into loops and accumulating Berry phases. The adiabatic error cancellation established in Theorem 1 applies to each loop individually, and may therefore reduce the total cost of Chern number computation. However, the interplay between the discretization of the surface and the adiabatic error requires careful analysis.

Second, when the ground state is degenerate, adiabatic transport around a loop produces a $U(d)$ holonomy matrix (the Wilczek–Zee holonomy) rather than a scalar phase. The wave operator formalism adopted in this work extends naturally to degenerate spectral subspaces via the intertwining property $U_A(s)P_0 = P(s)U_A(s)$, where $P(s)$ is the spectral projection onto the degenerate subspace. Analyzing whether the forward–reverse error cancellation persists for the non-abelian case, and developing efficient algorithms for estimating the Wilczek–Zee holonomy, are promising directions for future work.

C. Comparing with the energy estimation

It is instructive to compare the present cancellation mechanism with the runtime-scaling method of Ref. [1], which we use only for branch resolution. Although that method is highly effective for determining the correct branch of the Berry phase, it does not exhibit the same

favorable error cancellation as the forward–reverse protocol developed here. The reason is fundamentally that the runtime-scaling method is based on energy estimation rather than Berry-phase estimation. More concretely, the cancellation mechanism in our protocol relies on the geometric nature of the Berry phase. By combining the forward adiabatic evolution under H with the reverse evolution under $-H$ along the same loop, the leading non-geometric contribution to the accumulated phase cancels, while the Berry phase itself is preserved. This structure is special to geometric phase estimation. In contrast, the runtime-scaling method of Ref. [1] extracts phase information through the scaling of the total accumulated phase with runtime, and is therefore sensitive to the dynamical-energy contribution itself. Since that procedure effectively performs energy estimation, its leading finite-time errors inherit the structure of energy-estimation errors and do not enjoy the same geometric cancellation.

A key conceptual insight emerging from this work is the fundamental difference between Berry phase estimation and usual ground energy estimation. In energy estimation, QPE on e^{-iHt} achieves Heisenberg-limited cost $O(1/\varepsilon)$: the eigenphase Et grows linearly with t , so the evolution time cancels between the signal and the precision requirement. In Berry phase estimation, θ_B is *intensive* in T —it does not grow with the adiabatic runtime—so increasing T only suppresses the systematic error without amplifying the signal. The adiabatic runtime thus appears as irreducible overhead, and the total cost $O(T/\varepsilon_B)$ necessarily exceeds the Heisenberg limit by a factor of T . Our error cancellation partially mitigates this gap by reducing the required runtime from $T = O(1/\varepsilon_B)$ to $T = O(1/\sqrt{\varepsilon_B})$, and Richardson extrapolation further improves the prefactor.

Although Berry phase estimation incurs an additional adiabatic runtime cost compared to energy estimation, the total computational cost should be considered together with the cost of ground state preparation. In energy estimation, the ground state is typically assumed to be provided as input (the guided local Hamiltonian setting), and the problem is BQP-complete [1]; without the guiding state, preparing the ground state is itself a hard problem (QMA-complete in general). In Berry phase estimation, the situation may be more favorable: since the quantity of interest is the geometric phase acquired along a trajectory of Hamiltonians, the initial ground state may be a simple structure that allows efficient preparation. A detailed analysis of the total complexity including state preparation cost for specific applications is an interesting direction for future work.

D. Combining with Early-FTQC Algorithms

An important direction for future work is to combine the present approach with recently proposed algorithms tailored to the early fault-tolerant regime [37–40], as well

as with Hamiltonian-simulation methods based on randomization and extrapolation, including multi-product-formula approaches [41, 42]. Our results suggest that Berry phase estimation is particularly well suited to this setting: the forward–reverse structure suppresses the dominant adiabatic error, Richardson extrapolation removes the leading non-oscillatory residual, and the large shot budgets naturally available in sampling-based early-FTQC algorithms can simultaneously support runtime randomization and statistical estimation.

This raises the possibility that Berry phases, and perhaps geometric quantities more broadly, may provide especially favorable targets for early fault-tolerant quantum advantage. It would therefore be interesting to study how the present error-suppression mechanism can be incorporated into concrete early-FTQC algorithmic frameworks.

DATA AVAILABILITY

The simulation code used in this work is available at <https://github.com/CKiumi/berry-phase-cancellation>.

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- [1] R. Hayakawa, K. Sakamoto, and C. Kiumi, Computational complexity of berry phase estimation in topological phases of matter (2025), arXiv:2509.13423 [quant-ph].
 - [2] R. P. Feynman, Simulating physics with computers, *International Journal of Theoretical Physics* **21**, 467 (1982).
 - [3] S. Lloyd, Universal quantum simulators, *Science* **273**, 1073 (1996).
 - [4] I. M. Georgescu, S. Ashhab, and F. Nori, Quantum simulation, *Reviews of Modern Physics* **86**, 153 (2014).
 - [5] J. Preskill, Quantum computing in the NISQ era and beyond, *Quantum* **2**, 79 (2018).
 - [6] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, and A. Kandala, Evidence for the utility of quantum computing before fault tolerance, *Nature* **618**, 500 (2023).
 - [7] M. V. Berry, Quantal phase factors accompanying adiabatic changes, *Proceedings of the Royal Society of London. A* **392**, 45 (1984).
 - [8] B. Simon, Holonomy, the quantum adiabatic theorem, and berry’s phase, *Physical Review Letters* **51**, 2167 (1983).
 - [9] M. Z. Hasan and C. L. Kane, Colloquium: Topological insulators, *Reviews of Modern Physics* **82**, 3045 (2010).
 - [10] X.-L. Qi and S.-C. Zhang, Topological insulators and superconductors, *Reviews of Modern Physics* **83**, 1057 (2011).
 - [11] R. D. King-Smith and D. Vanderbilt, Theory of polarization of crystalline solids, *Physical Review B* **47**, 1651 (1993).
 - [12] R. Resta, Manifestations of Berry’s phase in molecules and condensed matter, *Journal of Physics: Condensed Matter* **12**, R107 (2000).
 - [13] D. Vanderbilt, *Berry Phases in Electronic Structure Theory* (Cambridge University Press, 2018).
 - [14] N. Nagaosa, J. Sinova, S. Onoda, A. H. MacDonald, and N. P. Ong, Anomalous Hall effect, *Reviews of Modern Physics* **82**, 1539 (2010).
 - [15] D. Xiao, M.-C. Chang, and Q. Niu, Berry phase effects on electronic properties, *Reviews of Modern Physics* **82**, 1959 (2010).
 - [16] S. Peotta and P. Törmä, Superfluidity in topologically nontrivial flat bands, *Nature Communications* **6**, 8944 (2015).
 - [17] P. Törmä, S. Peotta, and B. A. Bernevig, Superconductivity, superfluidity and quantum geometry in twisted multilayer systems, *Nature Reviews Physics* **4**, 528 (2022).
 - [18] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, Quantized Hall conductance in a two-dimensional periodic potential, *Physical Review Letters* **49**, 405 (1982).
 - [19] F. Wilczek and A. Zee, Appearance of gauge structure in simple dynamical systems, *Physical Review Letters* **52**, 2111 (1984).
 - [20] P. Zanardi and M. Rasetti, Holonomic quantum computation, *Physics Letters A* **264**, 94 (1999).
 - [21] E. Sjöqvist, D. M. Tong, L. M. Andersson, B. Hessmo, M. Johansson, and K. Singh, Non-adiabatic holonomic quantum computation, *New Journal of Physics* **14**, 103035 (2012).
 - [22] T. Fukui, Y. Hatsugai, and H. Suzuki, Chern numbers in discretized Brillouin zone: Efficient method of computing (spin) Hall conductances, *Journal of the Physical Society of Japan* **74**, 1674 (2005).
 - [23] R. Yu, X. L. Qi, A. Bernevig, Z. Fang, and X. Dai, Equivalent expression of $\mathbb{Z}_2$ topological invariant for band insulators using the non-Abelian Berry connection, *Physical Review B* **84**, 075119 (2011).
 - [24] A. A. Soluyanov and D. Vanderbilt, Computing topological invariants without inversion symmetry, *Physical Review B* **83**, 235401 (2011).
 - [25] C. Zhang, Z. Wei, and A. Papageorgiou, Adiabatic quantum counting by geometric phase estimation, *Quantum Information Processing* **9**, 369 (2010).
 - [26] B. Murta, G. Catarina, and J. Fernández-Rossier, Berry phase estimation in gate-based adiabatic quantum simulation, *Physical Review A* **101**, 020302 (2020).
 - [27] S. Tamiya, S. Koh, and Y. O. Nakagawa, Calculating nonadiabatic couplings and Berry’s phase by variational quantum eigensolvers, *Physical Review Research*

- 3, 023244 (2021).
- [28] M. Mootz and Y.-X. Yao, Efficient Berry phase calculation via adaptive variational quantum computing approach, *APL Quantum* **3**, 016111 (2026).
 - [29] M. Born and V. Fock, Beweis des adiabatensatzes, *Zeitschrift fuer Physik* **51**, 165 (1928).
 - [30] T. Kato, On the adiabatic theorem of quantum mechanics, *Journal of the Physical Society of Japan* **5**, 435 (1950).
 - [31] M. V. Berry, Quantum phase corrections from adiabatic iteration, *Proceedings of the Royal Society of London. A* **414**, 31 (1987).
 - [32] Y. Aharonov and J. Anandan, Phase change during a cyclic quantum evolution, *Physical Review Letters* **58**, 1593 (1987).
 - [33] J. E. Avron, R. Seiler, and L. G. Yaffe, Adiabatic theorems and applications to the quantum hall effect, *Communications in Mathematical Physics* **110**, 33 (1987).
 - [34] G. Nenciu, Linear adiabatic theory. exponential estimates, *Communications in Mathematical Physics* **152**, 479 (1993).
 - [35] S. Jansen, M.-B. Ruskai, and R. Seiler, Bounds for the adiabatic approximation with applications to quantum computation, *Journal of Mathematical Physics* **48**, 102111 (2007).
 - [36] G. Rigolin, G. Ortiz, and V. H. Ponce, Beyond the quantum adiabatic approximation: Adiabatic perturbation theory, *Physical Review A* **78**, 052508 (2008).
 - [37] E. Granet and H. Dreyer, Hamiltonian dynamics on digital quantum computers without discretization error, *npj Quantum Information* **10**, 10.1038/s41534-024-00877-y (2024).
 - [38] C. Kiumi and B. Koczor, TE-PAI: Exact time evolution by sampling random circuits, *Quantum Science and Technology* **10**, 045071 (2025).
 - [39] H. Pages, C. Kiumi, Y. Morohoshi, B. Koczor, and K. Mitarai, Low-resource quantum energy gap estimation via randomization, *arXiv:2601.13881* (2026).
 - [40] T. Hayata and Y. Kikuchi, Continuous-time evolution via probabilistic angle interpolation and its applications (2026).
 - [41] A. Carrera Vazquez, D. J. Egger, D. Ochsner, and S. Woerner, Well-conditioned multi-product formulas for hardware-friendly hamiltonian simulation, *Quantum* **7**, 1067 (2023).
 - [42] J. D. Watson and J. Watkins, Exponentially reduced circuit depths using trotter error mitigation, *PRX Quantum* **6**, 10.1103/kw39-yxq5 (2025).

Appendix A: Adiabatic Perturbation Theory

In this appendix we review the adiabatic perturbation theory (APT) developed by Rigolin, Ortiz, and Ponce [36], adapted to the notation of the main text. The APT provides systematic $1/T$ corrections to the adiabatic approximation via purely algebraic recursion relations, without recourse to differential equations.

Let $H(s)$ be a smooth family of Hamiltonians for $s \in [0, 1]$ with $H(0) = H(1)$. For each s , let $E_n(s)$ and $|n(s)\rangle$ denote the n -th instantaneous eigenvalue and eigenstate, with the ground state $|\psi(s)\rangle = |0(s)\rangle$ assumed non-degenerate for all s . We define

$$\Delta_{nm}(s) := E_n(s) - E_m(s), \quad M_{nm}(s) := \langle n(s) | \dot{m}(s) \rangle,$$

where the dot denotes d/ds . Note that $M_{nn}(s)$ is purely imaginary (since $\langle n(s) | n(s) \rangle = 1$ implies $\Re(M_{nn}) = 0$), and $M_{nm}(s) = -M_{mn}^*(s)$ for $n \neq m$. We also define the instantaneous dynamical and Berry phases for each eigenstate:

$$\theta_D^{(m)}(s) := T \int_0^s E_m(\sigma) d\sigma, \quad \theta_B^{(m)}(s) := i \int_0^s M_{mm}(\sigma) d\sigma,$$

and the relative Berry phase $\beta_{mn}(s) := \theta_B^{(m)}(s) - \theta_B^{(n)}(s)$. The evolved state $|\varphi(s)\rangle := U(s) |\psi(0)\rangle$ is expanded in the instantaneous eigenbasis as

$$|\varphi(s)\rangle = \sum_n a_n(s) |n(s)\rangle.$$

If degeneracies occur in the excited spectrum, we fix once and for all a smooth orthonormal instantaneous basis within each degenerate excited subspace, and all quantities below are understood with respect to that choice. Substituting into the Schrödinger equation $i|\dot{\varphi}(s)\rangle = TH(s)|\varphi(s)\rangle$ and projecting onto $\langle m(s)|$ yields the coupled equations

$$\dot{a}_m(s) = -iT E_m(s) a_m(s) - \sum_n a_n(s) M_{mn}(s). \quad (\text{A1})$$

To remove the explicit dynamical and Berry phases, we define

$$a_m(s) = e^{-i\theta_D^{(m)}(s)} e^{i\theta_B^{(m)}(s)} b_m(s). \quad (\text{A2})$$

Substituting this into (A1) yields

$$\dot{b}_m(s) = - \sum_{n \neq m} e^{-iT\omega_{nm}(s)} e^{i\beta_{nm}(s)} M_{mn}(s) b_n(s), \quad (\text{A3})$$

where

$$\omega_{nm}(s) := \int_0^s \Delta_{nm}(\sigma) d\sigma.$$

Thus, after extracting the dynamical and Berry phases, the remaining amplitudes $b_m(s)$ are coupled only through the off-diagonal adiabatic couplings $M_{mn}(s)$, modulated by rapidly oscillating phase factors. This motivates the adiabatic perturbation expansion.

We therefore expand the state as

$$|\Psi(s)\rangle = \sum_{p=0}^{\infty} T^{-p} |\Psi^{(p)}(s)\rangle, \quad (\text{A4})$$

with

$$|\Psi^{(p)}(s)\rangle = \sum_n e^{-i\theta_D^{(n)}(s)} e^{i\theta_B^{(n)}(s)} b_n^{(p)}(s) |n(s)\rangle, \quad (\text{A5})$$

where

$$b_n^{(p)}(s) = \sum_m e^{iT\omega_{nm}(s)} e^{-i\beta_{nm}(s)} b_{nm}^{(p)}(s).$$

Equivalently,

$$|\Psi(s)\rangle = \sum_{n,m} \sum_{p=0}^{\infty} T^{-p} e^{-i\theta_D^{(m)}(s)} e^{i\theta_B^{(m)}(s)} b_{nm}^{(p)}(s) |n(s)\rangle.$$

Substituting this ansatz into the Schrödinger equation, projecting onto $\langle k(s)|$, and matching powers of $1/T$, we obtain

$$i\Delta_{nm}(s) b_{nm}^{(p+1)}(s) + \dot{b}_{nm}^{(p)}(s) + W_{nm}(s) b_{nm}^{(p)}(s) + \sum_{k \neq n} M_{nk}(s) b_{km}^{(p)}(s) = 0, \quad (\text{A6})$$

where

$$W_{nm}(s) := M_{nn}(s) - M_{mm}(s).$$

This recursion determines the coefficient $b_{nm}^{(p+1)}(s)$ at the next order in terms of the lower-order amplitudes. In this way, APT generates the adiabatic expansion systematically, with corrections organized in successive powers of $1/T$. The initial conditions at each order are fixed by requiring $|\Psi(0)\rangle = |\psi(0)\rangle$, which gives

$$b_{nm}^{(0)}(0) = b_n(0) \delta_{nm}, \quad b_n^{(p)}(0) = \sum_{m \neq n} b_{nm}^{(p)}(0) + b_{nn}^{(p)}(0) = 0 \quad (p \geq 1). \quad (\text{A7})$$

For a system starting in the ground state, $b_n(0) = \delta_{n0}$. At zeroth order, the recursion relation with $p = -1$ (or equivalently, the leading term of the ansatz) gives simply

$$b_{nm}^{(0)}(s) = b_n(0) \delta_{nm}. \quad (\text{A8})$$

Zero-th order.

The zero-th order solution yields the standard adiabatic approximation:

$$|\Psi^{(0)}(s)\rangle = \sum_n e^{-i\theta_D^{(n)}(s)} e^{i\theta_B^{(n)}(s)} b_n(0) |n(s)\rangle.$$

First order.

Off-diagonal coefficients ($n \neq m$):

Setting $p = 0$ in the recursion (A6) and using $\dot{b}_{nm}^{(0)} = 0$, $W_{nm} b_{nm}^{(0)} = 0$ (since $b_{nm}^{(0)} \propto \delta_{nm}$ and $W_{nn} = 0$), we obtain

$$b_{nm}^{(1)}(s) = \frac{iM_{nm}(s)}{\Delta_{nm}(s)} b_m(0), \quad n \neq m. \quad (\text{A9})$$

Diagonal coefficients ($n = m$):

When $n = m$, the term $i\Delta_{nn}(s)b_{nn}^{(p+1)}(s) = 0$ vanishes identically, so the recursion at order $p = 0$ becomes an identity. Instead, one uses the recursion at $p = 1$ with $n = m$:

$$\dot{b}_{nn}^{(1)}(s) + \sum_{k \neq n} M_{nk}(s) b_{kn}^{(1)}(s) = 0. \quad (\text{A10})$$

Integrating with (A9) and using $M_{nk} = -M_{kn}^*$ gives

$$b_{nn}^{(1)}(s) = i \sum_{m \neq n} J_{mn}(s) b_n(0) + b_{nn}^{(1)}(0), \quad (\text{A11})$$

where

$$J_{mn}(s) := \int_0^s \frac{|M_{mn}(\sigma)|^2}{\Delta_{mn}(\sigma)} d\sigma, \quad (\text{A12})$$

and the initial condition (A7) determines

$$b_{nn}^{(1)}(0) = -i \sum_{m \neq n} \frac{M_{nm}(0)}{\Delta_{nm}(0)} b_m(0). \quad (\text{A13})$$

First-order correction to the state:

Assembling these results for a general initial state

$$|\Psi(0)\rangle = \sum_m b_m(0) |m(0)\rangle,$$

the first-order correction is

$$\begin{aligned} |\Psi^{(1)}(s)\rangle = i \sum_{n \neq m} \left[e^{-i\theta_D^{(n)}(s)} e^{i\theta_B^{(n)}(s)} J_{mn}(s) b_n(0) \right. \\ \left. + \left(\frac{e^{-i\theta_D^{(m)}(s)} e^{i\theta_B^{(m)}(s)} M_{nm}(s)}{\Delta_{nm}(s)} - \frac{e^{-i\theta_D^{(n)}(s)} e^{i\theta_B^{(n)}(s)} M_{nm}(0)}{\Delta_{nm}(0)} \right) b_m(0) \right] |n(s)\rangle. \end{aligned} \quad (\text{A14})$$

The term proportional to $J_{mn}(s)$ gives a correction to the amplitude of each initial component, which is absent in the standard non-APT treatment. The remaining two terms describe couplings between different instantaneous eigenstates: the term proportional to $M_{nm}(s)/\Delta_{nm}(s)$ is non-oscillatory, while the term proportional to $M_{nm}(0)/\Delta_{nm}(0)$ carries the dynamical phase of the n -th level and is oscillatory.

Second order.

Off-diagonal coefficients ($n \neq m$).

Setting $p = 1$ and $n \neq m$ in the recursion (A6), using the first-order results, and separating the $k = m$ term from the sum yields

$$b_{nm}^{(2)}(s) = \frac{i}{\Delta_{nm}(s)} \left[i \frac{d}{ds} \left(\frac{M_{nm}(s)}{\Delta_{nm}(s)} \right) b_m(0) + W_{nm} b_{nm}^{(1)}(s) + M_{nm}(s) b_{mm}^{(1)}(s) + \sum_{k \neq n, m} M_{nk}(s) b_{km}^{(1)}(s) \right]$$

Substituting the first-order results (A9)–(A11) and simplifying gives

$$\begin{aligned} b_{nm}^{(2)}(s) = -\frac{1}{\Delta_{nm}(s)} \left[\frac{d}{ds} \left(\frac{M_{nm}(s)}{\Delta_{nm}(s)} \right) + \frac{W_{nm}(s) M_{nm}(s)}{\Delta_{nm}(s)} \right. \\ \left. + M_{nm}(s) \sum_{k \neq m} J_{km}(s) + \sum_{\substack{k \neq n \\ k \neq m}} \frac{M_{nk}(s) M_{km}(s)}{\Delta_{km}(s)} \right] b_m(0) \\ + \frac{M_{nm}(s)}{\Delta_{nm}(s)} \sum_{k \neq m} \frac{M_{mk}(0)}{\Delta_{mk}(0)} b_k(0). \end{aligned} \quad (\text{A15})$$

Diagonal coefficients ($n = m$): Setting $p = 2$ and $n = m$ in the recursion gives

$$b_{nn}^{(2)}(s) = - \int_0^s \sum_{m \neq n} M_{nm}(\sigma) b_{nm}^{(2)}(\sigma) d\sigma + b_{nn}^{(2)}(0), \quad (\text{A16})$$

where $b_{nn}^{(2)}(0) = - \sum_{m \neq n} b_{nm}^{(2)}(0)$ from (A7), and $b_{nm}^{(2)}(0)$ is obtained from (A15) at $s = 0$.

Second-order correction to the state: The full second-order correction is

$$|\Psi^{(2)}(s)\rangle = \sum_{n,m} e^{-i\theta_D^{(m)}(s)} e^{i\theta_B^{(m)}(s)} b_{nm}^{(2)}(s) |n(s)\rangle, \quad (\text{A17})$$

with the coefficients given by (A15) and (A16). A key feature of the APT is that all corrections at order p are determined algebraically from the coefficients at order $p - 1$, requiring only the instantaneous eigenvalues $E_n(s)$ and eigenstates $|n(s)\rangle$ of $H(s)$.

Appendix B: APT analysis of the adiabatic phase error

In this appendix we use the adiabatic perturbation theory reviewed in Appendix A to derive the phase error expansion of Lemma 1. The scalar adiabatic error amplitude $z(s) = e^{i\theta_D(s)} e^{-i\theta_B(s)} \langle \psi(s) | U_T(s) | \psi(0) \rangle$ is obtained directly from the APT ansatz (A5): projecting $|\Psi(s)\rangle$ onto the ground state $\langle 0(s) |$ and removing the ground-state dynamical and Berry phases gives

$$z(s) = \sum_m \sum_{p \geq 0} T^{-p} e^{-iT\omega_{m0}(s)} e^{i\beta_{m0}(s)} b_{0m}^{(p)}(s), \quad (\text{B1})$$

where $\omega_{m0}(s) = \int_0^s \Delta_{m0}$, the $b_{0m}^{(p)}(s)$ are the T -independent coefficients of the recursion (A6), and we used the ground-state initial condition $b_n(0) = \delta_{n0}$. The $m = 0$ terms are non-oscillatory and start from $b_{00}^{(0)}(s) = 1$, since the zeroth-order amplitudes are $b_{0m}^{(0)} = \delta_{0m}$. Writing $z = 1 + \delta$, the error amplitude splits into non-oscillatory ($m = 0$) and oscillatory ($m \neq 0$) parts, $\delta = \delta_{\text{bulk}} + \delta_{\text{osc}}$, where

$$\delta_{\text{bulk}}(s) = \sum_{p \geq 1} \frac{b_{00}^{(p)}(s)}{T^p}, \quad \delta_{\text{osc}}(s) = \sum_{p \geq 1} \frac{1}{T^p} \sum_{m \neq 0} e^{-iT\omega_{m0}(s)} e^{i\beta_{m0}(s)} b_{0m}^{(p)}(s). \quad (\text{B2})$$

The phase error $\varphi(s) := \arg z(s)$ is the imaginary part of the principal logarithm,

$$\varphi(s) = \Im \log(1 + \delta(s)) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \Im(\delta(s)^k), \quad (\text{B3})$$

convergent for $|\delta(s)| < 1$. Factoring $1 + \delta = (1 + \delta_{\text{bulk}})(1 + \delta_{\text{osc}}/(1 + \delta_{\text{bulk}}))$ before taking the logarithm separates the geometric and oscillatory contributions, $\varphi = \varphi_{\text{bulk}} + \varphi_{\text{osc}}$, with

$$\varphi_{\text{bulk}} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \Im(\delta_{\text{bulk}}^k), \quad \varphi_{\text{osc}} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \Im \left[\left(\frac{\delta_{\text{osc}}}{1 + \delta_{\text{bulk}}} \right)^k \right].$$

The first sum is built solely from the geometric coefficients $b_{00}^{(p)}$ and carries no T -oscillation; collecting powers of $1/T$ gives the bulk coefficients φ_k . In the second sum, every term carries the factor δ_{osc}^k with $k \geq 1$, hence at least one $e^{-iT\omega_{m0}}$, and gives the boundary coefficients $\varphi_k^{(T)}$. Since the expansion contains only powers of $\delta_{\text{osc}}/(1 + \delta_{\text{bulk}})$ and never a conjugate, each oscillatory term retains a net frequency $\Omega \geq \Delta_{\text{min}}$ given by a positive sum of gaps. Because $b_{0m}^{(1)} = 0$ for $m \neq 0$, the oscillatory sum is $O(T^{-2})$, so the leading boundary term is $\varphi_2^{(T)}$.

More generally, the oscillatory coefficients have the following form. For each fixed order k , there exists a finite index set I_k such that

$$\varphi_k^{(T)} = \sum_{\nu \in I_k} [a_{k,\nu} \cos(\Omega_{k,\nu} T) + b_{k,\nu} \sin(\Omega_{k,\nu} T)], \quad (\text{B4})$$

where $a_{k,\nu}$ and $b_{k,\nu}$ are independent of T . Each frequency $\Omega_{k,\nu}$ is a positive sum of integrated gaps,

$$\Omega_{k,\nu} = \omega_{n_1 0}(1) + \cdots + \omega_{n_q 0}(1), \quad q \geq 1, \quad (\text{B5})$$

and hence satisfies

$$\Omega_{k,\nu} \geq \Delta_{\min} > 0.$$

Thus no zero-frequency component appears in the oscillatory sector.

First order. At first order the recursion (A6) with the ground-state initial condition gives the non-oscillatory coefficient

$$b_{00}^{(1)}(s) = i \sum_{m \neq 0} J_{m0}(s), \quad (\text{B6})$$

while the off-diagonal survival coefficients vanish, $b_{0m}^{(1)}(s) = 0$ for $m \neq 0$. Hence at order $1/T$ the oscillatory part δ_{osc} contributes nothing, and from (B3),

$$\varphi_1(s) = \Im(b_{00}^{(1)}(s)) = \int_0^s \sum_{n \neq 0} \frac{|M_{n0}(\sigma)|^2}{\Delta_{n0}(\sigma)} d\sigma, \quad \varphi_1^{(T)}(s) = 0. \quad (\text{B7})$$

The vanishing of $b_{0m}^{(1)}$ is the reason there is no first-order boundary term; the oscillatory contribution first appears at second order, through $b_{0m}^{(2)}$, as $\varphi_2^{(T)}$.

Second order. At second order both the bulk and oscillatory coefficients contribute. From (B3), collecting the $1/T^2$ terms of $\Im \log(1 + \delta)$ gives

$$\frac{\varphi_2 + \varphi_2^{(T)}}{T^2} = \frac{1}{T^2} \Im \left(b_{00}^{(2)} - \frac{1}{2} (b_{00}^{(1)})^2 \right) + \frac{1}{T^2} \sum_{m \neq 0} \Im \left(e^{-iT\omega_{m0}} e^{i\beta_{m0}} b_{0m}^{(2)} \right),$$

since $b_{0m}^{(1)} = 0$ removes the $m \neq 0$ contribution to δ_{osc} at first order. The first group is non-oscillatory and gives φ_2 ; the second carries $e^{-iT\omega_{m0}}$ and gives $\varphi_2^{(T)}$.

Bulk coefficient. The ground-state coefficient $b_{00}^{(2)}(s)$ follows from the recursion (A6) with $n = 0$. Using $W_{n0} = M_{nn} - M_{00} = -i(\dot{\theta}_B^{(n)} - \dot{\theta}_B)$ and writing $M_{n0} = |M_{n0}| e^{i\phi_n}$, so that $\Im(\dot{M}_{n0} M_{n0}^*) = \dot{\phi}_n |M_{n0}|^2$, the imaginary part evaluates to

$$\varphi_2(s) = \sum_{n \neq 0} \int_0^s \frac{(\dot{\beta}_{n0}(\sigma) - \dot{\phi}_n(\sigma)) |M_{n0}(\sigma)|^2}{\Delta_{n0}(\sigma)^2} d\sigma - i \int_0^s \mathcal{A}^{(2)}(\sigma) d\sigma, \quad (\text{B8})$$

where $\dot{\beta}_{n0} = \dot{\theta}_B^{(n)} - \dot{\theta}_B$ and

$$\mathcal{A}^{(2)}(s) := \sum_{\substack{n,k \neq 0 \\ n \neq k}} \frac{M_{0n}(s) M_{nk}(s) M_{k0}(s)}{\Delta_{n0}(s) \Delta_{k0}(s)}$$

is purely imaginary, so that $-i \int_0^s \mathcal{A}^{(2)} d\sigma$ is real, as follows from the anti-hermiticity relation $M_{ab}^* = -M_{ba}$ and relabeling the summation indices.

Oscillatory coefficient. The off-diagonal coefficient $b_{0m}^{(2)}(s)$ inherits the endpoint factor $M_{m0}(0)/\Delta_{m0}(0)$ from the first-order excited amplitude. Substituting into the oscillatory group above and writing $M_{m0} = |M_{m0}| e^{i\phi_m}$ yields

$$\varphi_2^{(T)}(s) = - \sum_{m \neq 0} \frac{|M_{m0}(0)| |M_{m0}(s)|}{\Delta_{m0}(s) \Delta_{m0}(0)} \sin(\beta_{m0}(s) + \phi_m(0) - \phi_m(s) - T\omega_{m0}(s)). \quad (\text{B9})$$

At the endpoint $s = 1$, with the closed loop $\Delta_{m0}(1) = \Delta_{m0}(0)$, this reduces to the form stated in Lemma 1.

Appendix C: Bound on the second-order term

Here we provide the details of the bound for leading term appearing in Theorem 1. To bound $\tilde{\theta}_B(T) - \theta_B$ in terms of physical parameters, we estimate each term separately.

a. Term 1: the $(\dot{\beta}_{n0} - \dot{\phi}_n)$ integral. This term involves $M_{n0}^* b_{n0}^{(2)}$ before splitting into sub-terms. From the explicit form of $b_{n0}^{(2)}$ in (A15), each contribution is bounded by products of $|M|$'s, $|\dot{M}|$'s, and inverse gaps. Differentiating $M_{n0} = \langle n | \dot{H} | \psi \rangle / \Delta_{n0}$ gives

$$|\dot{M}_{n0}| \leq \frac{\ddot{H}_{\max}}{\Delta_{\min}} + \frac{\dot{H}_{\max}^2}{\Delta_{\min}^2}.$$

Combining with other factor, the integrand satisfies

$$\left| \frac{(\dot{\beta}_{n0} - \dot{\phi}_n) |M_{n0}|^2}{\Delta_{n0}^2} \right| \leq \frac{\dot{H}_{\max}^2}{\Delta_{\min}^4} \left(\frac{\ddot{H}_{\max}}{\Delta_{\min}} + \frac{\dot{H}_{\max}^2}{\Delta_{\min}^2} \right).$$

b. Term 2: the $\mathcal{A}^{(2)}$ integral. Introduce the vector $v(s)$ on the excited subspace by

$$v_n(s) := \frac{M_{n0}(s)}{\Delta_{n0}(s)}, \quad n \neq 0.$$

Then $\mathcal{A}^{(2)}$ can be viewed as the quadratic form

$$\mathcal{A}^{(2)}(s) = \sum_{\substack{n,k \neq 0 \\ n \neq k}} M_{0n}(s) M_{nk}(s) v_k(s) = - \sum_{\substack{n,k \neq 0 \\ n \neq k}} \overline{v_n(s)} M_{nk}(s) v_k(s),$$

where we used $M_{0n} = -\overline{M_{n0}}$. Hence

$$|\mathcal{A}^{(2)}(s)| \leq \|M_Q(s)\| \|v(s)\|^2,$$

where $M_Q(s) = (M_{nk}(s))_{n,k \neq 0}$ denotes the excited-sector Berry-connection matrix. Now

$$\|v(s)\|^2 = \sum_{n \neq 0} \frac{|M_{n0}(s)|^2}{\Delta_{n0}(s)^2} \leq \frac{\dot{H}_{\max}^2}{\Delta_{\min}^4},$$

using $|M_{n0}(s)| \leq \dot{H}_{\max}/\Delta_{\min}$. If we further introduce the minimum excited-state splitting

$$\gamma_{\text{ex}} := \min_{s \in [0,1]} \min_{\substack{n \neq k \\ n,k \neq 0}} |E_n(s) - E_k(s)|,$$

then the Hellmann–Feynman relation gives

$$\|M_Q(s)\| \leq \frac{\dot{H}_{\max}}{\gamma_{\text{ex}}}.$$

Therefore

$$|\mathcal{A}^{(2)}(s)| \leq O\left(\frac{\dot{H}_{\max}^3}{\Delta_{\min}^4 \gamma_{\text{ex}}}\right),$$

and hence

$$\left| \int_0^1 \mathcal{A}^{(2)}(s) ds \right| \leq O\left(\frac{\dot{H}_{\max}^3}{\Delta_{\min}^4 \gamma_{\text{ex}}}\right).$$

When degeneracies or level crossings occur within the excited spectrum, the level-by-level quantities M_{nk} are no longer the most natural objects, since the relevant geometry is that of the excited manifold as a whole rather than of individual excited states. Accordingly, we expect that the term $\mathcal{A}^{(2)}$ should be reformulated in terms of the Berry connection on the excited manifold, while the overall Berry-phase error scaling should remain unchanged as long as the ground state stays non-degenerate and separated from the excited manifold by a nonzero gap.

c. Term 3: the oscillatory boundary term. Using $|\sin(\cdot)\cos(\cdot)| \leq 1$ and $|M_{n0}(0)|, |M_{n0}(1)| \leq \dot{H}_{\max}/\Delta_{\min}$:

$$|\Phi_2^{(T)}| \leq \frac{\dot{H}_{\max}^2}{\Delta_{\min}^4}.$$

d. Combined bound. Summing over excited states (at most N_{eff} levels couple appreciably to the ground state) and integrating over $s \in [0, 1]$, we obtain

$$|\tilde{\theta}_B(T) - \theta_B| \leq \frac{C_2}{T^2}, \quad (\text{C1})$$

$$C_2 = O\left(\frac{\dot{H}_{\max}^2 \ddot{H}_{\max}}{\Delta_{\min}^5} + \frac{\dot{H}_{\max}^4}{\Delta_{\min}^6} + \frac{\dot{H}_{\max}^3}{\Delta_{\min}^4 \gamma_{\text{ex}}}\right).$$

We note that the last contribution, proportional to γ_{ex}^{-1} , is the only part of the present bound that is sensitive to degeneracies or near-degeneracies within the excited spectrum. However, as discussed below, this term is eliminated by Richardson extrapolation, so the final complexity bound is unaffected by the detailed structure of the excited-state splittings.

Appendix D: Properties of the Richardson factor

We collect elementary properties of the factor

$$L_m(\alpha) := \sum_{\ell=0}^m |w_{m,\ell}| \alpha^{-2\ell}, \quad \alpha > 1, \quad (\text{D1})$$

which appears in the bound for the leading oscillatory sector. This quantity should not be confused with the statistical amplification factor of the Richardson estimator. Rather, $L_m(\alpha)$ is the effective coefficient factor for the T^{-2} oscillatory contribution.

More generally, the k -th oscillatory sector is controlled by

$$L_{m,k}(\alpha) := \sum_{\ell=0}^m |w_{m,\ell}| \alpha^{-k\ell}. \quad (\text{D2})$$

Indeed, the factor $\alpha^{-k\ell}$ comes from

$$(\alpha^\ell T)^{-k} = \alpha^{-k\ell} T^{-k}.$$

Thus

$$L_m(\alpha) = L_{m,2}(\alpha). \quad (\text{D3})$$

Moreover, for every $k \geq 2$,

$$L_{m,k}(\alpha) \leq L_m(\alpha), \quad (\text{D4})$$

because $\alpha^{-k\ell} \leq \alpha^{-2\ell}$. Hence $L_m(\alpha)$ also gives a uniform upper bound for all higher oscillatory sectors $k \geq 2$, although the exact factor for $k > 2$ is $L_{m,k}(\alpha)$.

The Richardson weights imply the product representation

$$L_m(\alpha) = \alpha^{-2m} \frac{\prod_{j=0}^{m-1} (1 + \alpha^{-2j})}{\prod_{j=1}^m (1 - \alpha^{-2j})}. \quad (\text{D5})$$

Consequently,

$$\frac{L_{m+1}(\alpha)}{L_m(\alpha)} = \alpha^{-2} \frac{1 + \alpha^{-2m}}{1 - \alpha^{-2m-2}}. \quad (\text{D6})$$

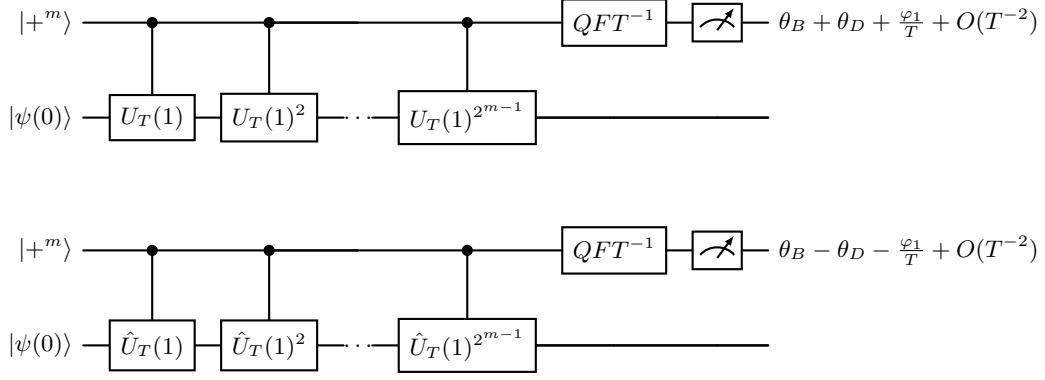


FIG. 5. Quantum circuit for Step 1 of the Berry phase estimation algorithm. Two independent QPE procedures are run on the forward propagator $U_T(1)$ and the reverse propagator $\hat{U}_T(1)$, each with input $|\psi(0)\rangle$. The upper circuit yields an estimate of the eigenphase $\theta_B + \theta_D + \varphi_1/T + O(T^{-2})$, while the lower circuit yields $\theta_B - \theta_D - \varphi_1/T + O(T^{-2})$. Adding the two outcomes cancels both the dynamical phase θ_D and the leading $O(T^{-1})$ phase error, leaving $2\theta_B + O(T^{-2})$.

Hence $L_m(\alpha)$ is monotone decreasing in m whenever

$$\alpha^{-2} + 2\alpha^{-4} \leq 1, \quad (\text{D7})$$

or equivalently

$$\alpha \geq \sqrt{2}. \quad (\text{D8})$$

Thus, for the common choice $\alpha = 2$, the effective coefficient of the leading oscillatory sector decreases with the Richardson order. For example, for $k = 2$,

$$L_{1,2}(2) = \frac{2}{3}, \quad L_{2,2}(2) = \frac{2}{9}, \quad L_{3,2}(2) = \frac{34}{567}. \quad (\text{D9})$$

For higher oscillatory sectors the factor is even smaller. For instance, for $k = 3$,

$$L_{1,3}(2) = \frac{1}{2}, \quad L_{2,3}(2) = \frac{1}{10}. \quad (\text{D10})$$

On the other hand, the crude estimate

$$L_m(\alpha) \leq (m+1) \left(\frac{\alpha^2}{\alpha^2 - 1} \right)^m \quad (\text{D11})$$

is only a worst-case upper bound. It does not capture the possible decrease of $L_m(\alpha)$ with m .

Appendix E: Berry phase estimation by Quantum Phase Estimation

We now show how the adiabatic error cancellation established in Theorem 1 and the Richardson extrapolation of Theorem 2 translate into an improved algorithm for estimating the Berry phase θ_B . We assume the setting of Section II, and that a lower bound on $\Delta_{\min}$ is known; this is a standard assumption in adiabatic quantum computation.

The ground state $|\psi(0)\rangle$ of $H(0)$ is given exactly as input. Given a target accuracy $\varepsilon_B > 0$ and failure probability $\eta \in (0, 1)$, the algorithm outputs an estimate $\hat{\theta}$ satisfying $|\hat{\theta} - \theta_B| \leq \varepsilon_B \pmod{2\pi}$ with probability at least $1 - \eta$. We allocate the failure budget as $\eta = \eta_{\text{precise}} + \eta_{\text{branch}}$, where η_{precise} is assigned to the high-precision estimation (Step 1) and η_{branch} to the coarse branch resolution (Step 2). For simplicity, we additionally assume $\dot{H}(0) = \dot{H}(1)$. This does not affect the essential conclusions, but only simplifies the endpoint-dependent prefactors in the T^{-2} error bounds.

Step 1: Forward–reverse estimation at two runtimes.

Run the forward–reverse protocol at two runtimes T_0 and αT_0 with $\alpha > 1$. For each runtime, perform QPE separately on U_T and $\tilde{U}_T$ with input $|\psi(0)\rangle$ (see Fig. 5), add the two outcomes mod 2π , and divide by two to obtain estimates $\tilde{\theta}_B(T_0)$ and $\tilde{\theta}_B(\alpha T_0)$ of $\theta_B \pmod{\pi}$. By Theorem 2, it suffices to choose

$$T_0 = O\left(\frac{\|\dot{H}(0)\|}{\Delta(0)^2 \sqrt{\varepsilon_B}}\right). \quad (\text{E1})$$

Choosing $m = O(\log(1/\varepsilon_B))$ precision qubits and $O(\log(1/\eta_{\text{precise}}))$ repetitions with majority vote ensures that the four QPE runs return the corresponding forward–reverse estimates $\tilde{\theta}_B(T_0)$ and $\tilde{\theta}_B(\alpha T_0)$ with the required phase-estimation accuracy for the subsequent branch-lifting and Richardson steps.

Each of the four QPE runs in this step can fail either because QPE returns an incorrect phase, with probability η_{QPE} , or because the evolution leaks out of the instantaneous ground-state sector. By Proposition 1, the latter occurs with probability

$$p_{\text{leak}}(1) \leq \frac{C_{\text{leak}}}{T_0^2} + O(T_0^{-3}) = O(\varepsilon_B),$$

for the above choice of T_0 . Hence, by a union bound over the four runs and a constant-factor increase of T_0 if needed, the overall success probability of Step 1 is at least $1 - \eta_{\text{precise}}$.

Step 2: Branch resolution via runtime scaling.

The estimates $\tilde{\theta}_B(T_0)$ and $\tilde{\theta}_B(\alpha T_0)$ determine θ_B only modulo π . To resolve this ambiguity, we employ the runtime-scaling technique of Ref. [1]. Run QPE separately on $U_{T_1}(1)$ and $U_{\alpha' T_1}(1)$ with input $|\psi(0)\rangle$, where

$$T_1 = O\left(\frac{\dot{H}_{\text{max}}^2}{\Delta_{\text{min}}^3}\right), \quad \alpha' = 1 + \frac{\pi}{T_1 H_{\text{max}} + \pi},$$

with $O(1)$ precision qubits and $O(\log(1/\eta_{\text{branch}}))$ repetitions. A coarse estimate $\hat{\theta}_B^{\text{coarse}}$ satisfying $|\hat{\theta}_B^{\text{coarse}} - \theta_B| \leq \pi/4$ is reconstructed as in Ref. [1]. The leakage probability $O(T_1^{-2})$ is a constant independent of ε_B and can be suppressed by increasing T_1 by a constant factor. By a union bound over the two QPE runs, the success probability of this step is at least $1 - \eta_{\text{branch}}$. The cost is $O(\dot{H}_{\text{max}}^2 \log(1/\eta_{\text{branch}})/\Delta_{\text{min}}^3)$, independent of ε_B .

Step 3: Branch lifting and Richardson extrapolation.

Using $\hat{\theta}_B^{\text{coarse}}$, lift each mod- π estimate to the unique representative in the interval

$$I := (\hat{\theta}_B^{\text{coarse}} - \pi/2, \hat{\theta}_B^{\text{coarse}} + \pi/2).$$

Concretely, let $\theta_B^{\text{lift}}(T_0)$ be the unique representative of $\tilde{\theta}_B(T_0) \pmod{\pi}$ lying in I , and similarly let $\theta_B^{\text{lift}}(\alpha T_0)$ be the unique representative of $\tilde{\theta}_B(\alpha T_0) \pmod{\pi}$ lying in I . Since both estimates are lifted into the same interval I of width π , their difference is represented on a common local branch, and the Richardson combination is therefore well defined.

We then form the Richardson extrapolant from these lifted values:

$$\hat{\theta}_{\text{sep}} := \frac{\alpha^2 \theta_B^{\text{lift}}(\alpha T_0) - \theta_B^{\text{lift}}(T_0)}{\alpha^2 - 1}.$$

Finally, output

$$\hat{\theta} := \hat{\theta}_{\text{sep}} \pmod{2\pi}.$$

By Theorem 2, the choice (E1) ensures that, conditioned on the success of the preceding QPE and branch-resolution steps, the final estimate satisfies

$$|\hat{\theta} - \theta_B| \leq \varepsilon_B \pmod{2\pi}.$$

By a union bound, all estimates are simultaneously correct with probability at least $1 - \eta$.

The dominant cost is Step 1, which requires four QPE runs (forward and reverse at runtimes T_0 and αT_0). Each QPE run uses $m = O(\log(1/\varepsilon_B))$ precision qubits and $O(\log(1/\eta_{\text{precise}}))$ repetitions with majority vote, for a total of $O(\log(1/\eta_{\text{precise}})/\varepsilon_B)$ controlled applications per run, each requiring Hamiltonian simulation of duration T_0 . By Proposition 1 and Theorem 2, the required runtime is

$$T_0 = O\left(\frac{\|\dot{H}(0)\|}{\Delta(0)^2 \sqrt{\varepsilon_B}}\right).$$

Step 2 costs $O(\dot{H}_{\max}^2 \log(1/\eta_{\text{branch}})/\Delta_{\min}^3)$ and is subdominant for small ε_B . The total Hamiltonian simulation time is therefore

$$O\left(\frac{\|\dot{H}(0)\|}{\Delta^2(0) \varepsilon_B^{3/2}} \cdot \log \frac{1}{\eta}\right). \quad (\text{E2})$$

Without Richardson extrapolation, the error bound of Theorem 1 requires the larger runtime, giving total cost derived in Appendix C as

$$O\left(\frac{\log(1/\eta)}{\varepsilon_B^{3/2}} \max\left(\frac{\dot{H}_{\max} \ddot{H}_{\max}^{1/2}}{\Delta_{\min}^{5/2}}, \frac{\dot{H}_{\max}^2}{\Delta_{\min}^3}, \frac{\dot{H}_{\max}^{3/2}}{\Delta_{\min}^2 \gamma_{\text{ex}}^{1/2}}\right)\right).$$

Richardson eliminates the $\ddot{H}_{\max}$ -dependence entirely, reduces the $\dot{H}_{\max}$ -dependence from quadratic to linear, and improves the gap dependence by one power of $\Delta_{\min}$, at the cost of two additional QPE runs. The ε_B -scaling remains $O(\varepsilon_B^{-3/2})$ in both cases.

This should be compared with two prior approaches. The runtime-scaling algorithm of Ref. [1] estimates θ_B over the full range $[0, 2\pi)$ by comparing adiabatic evolutions at two different runtimes, but does not employ forward–reverse cancellation. The systematic phase error is $O(T^{-1})$, requiring $T = O(\dot{H}_{\max}^2/(\Delta_{\min}^3 \varepsilon_B))$ and giving total cost $O(\dot{H}_{\max}^2 \log(1/\eta)/(\Delta_{\min}^3 \varepsilon_B^2))$. The algorithm of Ref. [26] cancels the dynamical phase by composing forward and backward evolutions into a single echo operator, but restricts the estimable range to $[0, \pi)$ and provides no analysis of the adiabatic phase error scaling or algorithmic complexity in terms of the Hamiltonian parameters. Our work provides the explicit phase error expansion (Lemma 1), proves the $O(T^{-2})$ cancellation (Theorem 1), further improves the prefactor via Richardson extrapolation (Theorem 2), achieves full $[0, 2\pi)$ range via branch resolution, and gives explicit complexity bounds in terms of $\dot{H}_{\max}$, $\ddot{H}_{\max}$, and $\Delta_{\min}$.